Honours Project: Bio-mathematics 747

Thesis

23 October 2025

Mechanistic modelling of metabolic switching in the human microbiota

Jemma Bennett

25136283

Supervisors: Dr Jack Jansma & Prof Cang Hui

UNIVERSITEIT·STELLENBOSCH·UNIVERSITY
jou kennisvennoot • your knowledge partner

Plagiaatverklaring / Plagiarism *Declaration*

1 Plagiaat is die oorneem en gebruik van die idees, materiaal en ander intellektuele eiendom van ander persone asof dit jou eie werk is.
Plagiarism is the use of ideas, material and other intellectual property of another's work and to present it as my own.

2 Ek erken dat die pleeg van plagiaat 'n strafbare oortreding is aangesien dit 'n vorm van diefstal is.
I agree that plagiarism is a punishable offence because it constitutes theft.

3 Ek verstaan ook dat direkte vertalings plagiaat is.
I also understand that direct translations are plagiarism.

4 Dienooreenkomstig is alle aanhalings en bydraes vanuit enige bron (ingesluit die internet) volledig verwys (erken). Ek erken dat die woordelikse aanhaal van teks sonder aanhalingstekens (selfs al word die bron volledig erken) plagiaat is.
Accordingly all quotations and contributions from any source whatsoever (including the internet) have been cited fully. I understand that the reproduction of text without quotation marks (even when the source is cited) is plagiarism.

5 Ek verklaar dat die werk in hierdie skryfstuk vervat, behalwe waar anders aangedui, my eie oorspronklike werk is en dat ek dit nie vantevore in die geheel of gedeeltelik ingehandig het vir bepunting in hierdie module/werkstuk of 'n ander module/werkstuk nie.
I declare that the work contained in this assignment, except where otherwise stated, is my original work and that I have not previously (in its entirety or in part) submitted it for grading in this module/assignment or another module/assignment.

25136283 **Studentenommer / *Student number***	**Handtekening / *Signature***
JL Bennett **Voorletters en van / *Initials and surname***	23 October 2025 **Datum / *Date***

Abstract

The human gut microbiota is a complex ecosystem whose interactions with metabolites are crucial for maintaining health. Understanding the dynamics of this system, particularly microbial strategies like metabolic switching, is experimentally challenging. This thesis utilises a mathematical framework to investigate the principles governing metabolite-bacteria interactions, their effect on community stability, and the emergence of complex network structures. A consumer-resource model based on ordinary differential equations (ODEs) was developed to simulate the dynamics of bacterial populations and three classes of metabolites: consumed, produced, and non-monotonic. Metabolic switching was incorporated via a Hill-type function for non-monotonic interactions. The model's behaviour was explored through a multi-faceted approach combining dynamical simulations, linear stability analysis, and network analysis for both small and large systems under closed (no external input) and open (constant metabolite influx) conditions. Simulations revealed that closed systems rapidly converge to a stable steady state, whereas open systems exhibit persistent, unstable dynamics characterized by oscillations. Stability analysis confirmed these findings, with the eigenvalues of the Jacobian matrix showing that external metabolite influx consistently drives instability. Network analysis demonstrated that interaction networks dynamically change over time, transitioning from dense, undifferentiated structures to sparse, specialised configurations at equilibrium. Furthermore, larger systems developed more complex network structures compared to smaller ones. Collectively, this research demonstrates that integrating dynamical simulations with stability and network analyses provides a powerful framework for exploring the ecological principles that shape gut microbiome organisation and resilience. These findings provide a theoretical foundation for understanding how microbial communities respond to perturbations and may inform the future design of targeted microbiota-based interventions to improve host health.

Table of Contents

1 Introduction 5

1.1 Key functions of gut microbiota in health 5

1.2 Non-monotonic metabolic behaviour 6

1.3 The use of mathematical models 8

1.4 Research question and objectives 9

2 Methods 10

2.1 Model implementation 11

2.2 Incorporating data 12

2.3 Simulations 13

2.4 Stability 15

2.5 Networks 16

3 Results 17

3.1 Baseline Dynamics in Closed Systems 17

3.2 Demonstration of Metabolic Switching 18

3.3 Influence of Nutrient Influx on System Dynamics 19

3.4 Stability Analysis 22

3.4.1 Eigenvalue spectra and stability classification 22

3.4.2 Bifurcation analysis 22

3.5 Network Structure and Temporal Evaluation 26

4 Discussion 30

4.1 Key findings 30

4.1.1 Simulations 30

4.1.2 Stability 32

4.1.3 Networks 34

4.2 Implications for microbiota interventions 38

4.3 Limitations and future work 39

5 Conclusion 41

6 References 42

7 Appendix 44

1 Introduction

The human gut microbiota is a microbial community inhabiting our gastrointestinal tract and includes bacteria, archaea, viruses and fungi [1]. These microbes engage in a symbiotic relationship with the host and can affect organs such as the gut, liver, muscle and brain [1]. The mutualistic relationship between host and microbes enhances nutrient absorption, strengthens the immune system and maintains homeostasis of the host, while the microbes benefit from the stable and nutrient-rich environment [2]. The microbial environment is driven by various complex interactions that take place during life such as diet, lifestyle, diseases and the use of antibiotics [3]. The collective genomes of the gut microbiota encode a vast array of enzymes that define its metabolic phenotype. This phenotype, in turn, directly alters the host's overall biochemistry [4]. Therefore, the gut microbiota plays a critical role in maintaining the health and function of the digestive system the overall homeostasis of the host [5]. A healthy gut microbiota is commonly characterised by a high microbial diversity [3]. The imbalance in gut microbiota and decrease in microbial diversity is linked to various conditions such as chronic inflammation and other physiological diseases, such as inflammatory bowel disease, cardiovascular disease, allergies, asthma and colorectal cancer [3]. Lately, there has been a notable increase in the number of cases of gut-related disorders in western populations associated with lifestyle-induced factors such as diet, including gastric ulcers caused by *H. Pylori* bacterium and metabolic diseases such as diabetes and obesity [6]. There is therefore much interest in the research into new therapeutic interventions that can manipulate the composition of the gut microbiota to benefit the health of the host [4].

1.1 Key functions of gut microbiota in health

The key functions of gut microbiota pertaining to health include digestion and production of beneficial metabolites, prevention of pathogen colonisation, immune system modulation as well as contributing to neurological and psychological health [7]. Many complex dietary carbohydrates, i.e. fiber, cannot be digested or absorbed by the human host and therefore reach the gut microbiota, which can metabolize them into short-chain fatty acids including acetate, propionate and butyrate which serve as an energy source for colonic cells and have various health benefits [8]. Besides serving

as an energy source, butyrate acts as an antimicrobial compound which is produced to create an environment that is not ideal for pathogens. Additionally, some microbes produce vitamins such as Vitamin K, which is essential for blood clotting and bone health, and B vitamins (e.g., B12, folate, and biotin), which play key roles in energy metabolism, DNA synthesis, and nervous system function, all of which are crucial for maintaining host health [9].

The gut environment, through interactions with the microbiota, plays a key role in the regulation and development of the immune system by affecting T-cell differentiation and antibody production, enhancing the body's defence mechanisms [10]. Recent studies suggest a very strong connection between the gut and the brain; this is known as the gut-brain axis [11]. The gut microbiota produces metabolites that interact with the central nervous system influencing brain function and behaviour. Consequently, imbalances in the gut microbiota have been associated with mental health disorders, including depression, anxiety and autism spectrum disorders. [11]

Microbes competitively inhibit growth of pathogenic bacteria by occupying ecological niches and producing antimicrobial compounds [9]. Nutrients and space are limited in the environment, and the beneficial microbes occupy the space and use the available nutrients. Therefore, there is no space for new, pathogenic bacteria to occupy the space, consume the resources and increase their population. The beneficial bacteria out-compete harmful bacteria, which protects the body and health is maintained. Additionally, the bacteria occupying the space produce a variety of antimicrobial agents which aid in defending their territory. These substances are toxic to pathogenic bacteria and can kill or inhibit their growth. Some examples include bacteriocins and butyrate [7].

1.2 Non-monotonic metabolic behaviour

A microorganism's metabolic strategy is defined by its capability of using nutrients to grow their population [7]. Microbes typically have multiple strategies to grow on multiple nutrients to survive in a variety of metabolic environments and which can involve metabolic switching [12]. The existence of these nutritional "preferences" has been known since Jacques Monod's work on diauxic shifts in the 1940s and 50s [12].

Escherichia coli is a classic example, which prefers to use glucose over other sugars when both are available. This preference extends to other sugars as well, with lactose, arabinose, xylose, sorbitol, rhamnose, and ribose being utilised in that specific order [12]. The choice between sequential nutrient use and simultaneous use is influenced by resource concentration, with simultaneous use being more common at low nutrient levels and sequential use at high concentrations [12]. Overflow metabolism is another strategy involving metabolic switching used by bacteria to optimise population growth in a nutritional dense environment [12]. This strategy is used by fast-growing cells where they opt for a less efficient energy production pathway, like fermentation, even when oxygen is available for the more efficient respiration pathway. This results in the excretion of metabolic byproducts, such as acetate, lactate, and ethanol, into the environment.[13] Flux Balance Analysis (FBA), a widely used computational method, has been able to successfully predict overflow metabolism and nutrient use preferences [14].

Gene-regulatory networks consist of transcription factors and elements that control the expression of genes in response to environmental signals, while metabolic networks represent the connection of biochemical reactions through which microbes process nutrients to generate energy and biomass. The dynamics between these two networks determine a microbe's "decision" to either co-utilise multiple carbon sources simultaneously or sequentially prioritise them, earlier defined as metabolic switching [12]. This coordination shows us how regulatory and metabolic networks work together for bacteria to implement optimal resource-use strategies [10]. Furthermore, because metabolite-bacteria interactions give rise to a highly interconnected network structure, these switching dynamics can also be analysed through the framework of network theory [15].

Another important aspect of microbial interactions is cross-feeding, a metabolic strategy where one species consumes the by-products of another. This process enables microbes to occupy complementary metabolic niches, reducing direct competition and fostering coexistence within a community [16]. In the gut microbiome, bacteria that ferment complex carbohydrates release acetate, which can be utilised by other species, ultimately benefiting both the microbial community and the host [17]. In

this way, cross-feeding exemplifies how metabolic strategies extend beyond individual survival to community-level stability and function.

Metabolic switching, overflow behaviour and cross-feeding are all examples of non-monotonic metabolite dynamics, as compounds may accumulate when production exceeds consumption and later declines as consumption increases.

1.3 The use of mathematical models

Due to the complex nature of metabolic networks and metabolic strategies of microbes within microbial communities, it is highly challenging to predict community behaviour. Consequently, making the manipulation using microbiota-targeted interventions to improve host health challenging. Although many studies investigate microbial communities using sequencing and metabolomics techniques, the dynamics of bacteria and metabolites are not well understood [18]. Mathematical models can simplify and describe microbial dynamics using equations which can be used to breach this gap and lay a theoretical foundation for the development of novel microbiota targeted interventions [18].

Collecting longitudinal data on gut microbiome composition and metabolite dynamics is experimentally challenging. Although stool samples can be collected at regular intervals, they typically provide daily measurements, whereas microbial composition and metabolite levels can fluctuate over much shorter timescales, often within hours [19] . Moreover, certain regions of the gastrointestinal tract, such as the small intestine, are difficult to access for direct sampling. As a result, in silico approaches offer a valuable alternative, allowing researchers to explore microbial and metabolic dynamics under controlled, repeatable conditions. Such simulations can overcome many of the practical limitations of experimental studies, given that the underlying model captures the key dynamic processes governing the system.

In silico sampling of microbial communities, which captures realistic interactions, has become an essential component of research in this field. However, theoretical models remain limited and challenging to develop [7]. The gut microbiome is a particularly challenging space, due to the overwhelming complexity of metabolite–bacteria

interactions. Models such as consumer-resource models make it possible to simulate scenarios, such as changing parameters regarding nutrient availability, and to investigate fundamental principles of microbial coexistence including metabolic switching strategies. They also enable extrapolation and interpolation of unobserved data, which can provide insights beyond what can be found experimentally [18]. For example, consumer-resource models have been used to explore how cross-feeding interactions stabilise community composition and how competition for limiting nutrients pushes exclusion versus coexistence [20]. Beyond these, generalised Lotka–Volterra (gLV) models have been widely applied to gut microbiome research; they have been used to infer microbial interaction networks from time-series data [21] and to predict how faecal microbiota transplantation perturbs microbial dynamics in recurrent *Clostridioides difficile* infection [22]. Agent-based models (ABMs), which incorporate spatial structure and stochasticity, have provided insight into how local metabolite gradients shape bacterial colonisation and how fluid flow in the colon influences microbial competition [23], [24]. Other frameworks such as FBA have been employed to simulate metabolic exchanges including studies that quantified how dietary changes affect microbial metabolism [25] and reconstructed metabolic models for hundreds of gut microbes to predict cross-feeding interactions [26]**.** Stochastic extensions of Lotka–Volterra models have further captured the role of random fluctuations in shaping microbial abundances and interaction networks [27]. Collectively, these approaches have enabled researchers to ask fundamental questions such as: *How do metabolic dependencies promote stability in complex gut ecosystems? Under what conditions does a rare taxon invade or go extinct? How do dietary shifts alter community structure over time?* By complementing experimental data, these models highlight the predictive power of mathematical abstraction while generating testable hypotheses that advance our understanding of the gut microbiome.

1.4 Research question and objectives

It is evident that the integrity and health of the gut microbiota are vital and closely linked to the overall health of the host [5]. Consequently, an unhealthy gut microbiota is associated with numerous chronic illnesses. This research therefore aims to mathematically describe the interactions between metabolites and bacteria in the gut to gain a deeper understanding of their dynamics, particularly the mechanisms

underlying metabolic switching. Ultimately, understanding these dynamics may reveal targets for manipulating microbial-metabolite interactions to improve host health. We hypothesise that mathematical modelling provides a viable and informative approach for investigating these processes.

Another aim of this study is to investigate how microbial and metabolite interactions shape the stability and behaviour of gut microbiome systems. Specifically, the research seeks to determine how different types of metabolites affect bacterial coexistence and system resilience under varying environmental conditions. By developing and analysing a mathematical model that captures key ecological and biochemical interactions, this study explores the mechanisms that promote or disrupt equilibrium within microbial communities and identifies factors that drive stability or instability in the system. We hypothesise that an open system, which is characterised by an influx of metabolites, portrays more instability than a closed system which is likely to show mostly stability. Together, this provides a comprehensive framework for understanding the dynamics and organisation of the microbiota.

2 Methods

To address these research questions, a multi-faceted modelling approach is used that combines dynamical simulations with network-based and stability analyses. First, both open and closed systems are simulated to capture how the presence or absence of external inputs influences metabolite dynamics and microbial interactions. The stability of these systems is then assessed to determine the conditions under which steady states are reached and how robust these equilibria are to perturbations. Additionally, network analysis techniques are applied to examine structural properties such as connectivity, centrality, and potential hubs of interaction. These analyses are conducted for both a small and large system. The small system allows for a clear interpretation of fundamental behaviours, and a larger system provides insight into the scalability and complexity of microbial-metabolite interactions. Together, these approaches provide a comprehensive framework for understanding the dynamics and organisation of the modelled microbiota systems.

2.1 Model implementation

Our model is based on a consumer-resource framework, where the consumers represent bacterial species and the resources represent metabolites [18]. The system describes the dynamic interactions between bacteria and metabolites in the gut microenvironment by a set of ordinary differential equations (ODEs) [28].

<u>Metabolites</u>

$$\frac{dM_i}{dt} = s_i[t] - M_i\left(\sum_{j\in\{B\}} c_{i,j}\cdot B_j\right) \tag{1}$$

<u>Bacteria</u>

$$\frac{dB_j}{dt} = B_j\left(-d_j + c_0\sum_{i\in\{M\}}\left(c_{i,j}^{+}\cdot M_i\right)\right)\left(k - \sum_{l\in\{B\}} B_l\right) \tag{2}$$

Equation (1) describes the rate of change in the concentration of a given metabolite, M_i. The concentration is influenced by the balance of two main processes, the influx of metabolites into the system denoted by $s_i[t]$ and the consumption / production of metabolites from the bacterial community. This rate is proportional to the metabolite's current concentration, M_i, and the sum of the influence of the bacterial levels. The interaction coefficient, $c_{i,j}$, describes the relationship between the given bacteria-metabolite pair. A negative $c_{i,j}$ means bacterium B_j produces a metabolite and a positive $c_{i,j}$ means bacterium B_j consumes the metabolite. Ultimately, $c_{i,j}$ describes the strength of the consumption of metabolite i by bacterium j.

Equation (2) describes the rate of change of a given bacterial population, B_j. The model is a form of logistic growth, where the overall growth is influenced by three factors: current population size, specific growth rate and carrying capacity. The current population size of a given bacteria population is B_j. The growth is proportional to the existing population. The specific growth rate term is $(-d_j + c_0\sum_{i\in\{M\}}(c_{i,j}^{+}\cdot M_i))$, and determines whether the population grows or shrinks. It consists of the constant death rate (d_j) and a growth term dependent on metabolite availability. Crucially, the growth term only sums over positive interactions ($c_{i,j}^{+}$) which means that bacteria only gain

biomass from metabolites that are taken up from the environment and $c_{i,j}^{+} = max(c_{i,j}, 0)$ [28]. The constant c_0 is the conversion coefficient that scales how efficiently metabolites are converted into new bacteria. The carrying capacity is the logistic term $(k - \sum B_l)$, which models the environmental limitations. As the total bacterial population $(\sum B_l)$ approaches the system's carrying capacity k, this term approaches zero, which slowly and eventually stops all growth due to factors such as limited space and resources.

Our model assumes a constant uptake rate, $c_{i,j}$, for a given bacterium-metabolite pair. To implement metabolites showing non-monotonic behaviour, a Hill-type metabolic switch can be incorporated into the interaction coefficient, $c_{i,j}$. Therefore, the uptake efficiency depends on metabolic availability which is a more realistic scenario for microbes with adaptive metabolism. This growth occurs where the effect of a metabolite on bacterial growth is not linear, and it switches at a certain threshold. Some metabolites may stimulate growth only above a certain level or have saturation effects, and thus a Hill function allows us to model the sigmoidal behaviour that occurs [29]. The interaction coefficient is depicted as:

$$c_{i,j}^{Non-mono} = c_{i,j}^{L} + \left(c_{i,j}^{U} - c_{i,j}^{L}\right) \frac{w_{i,j}^{K_{Hill}}}{w_{i,j}^{K_{Hill}} + \left(\frac{M_i}{B_j}\right)^{K_{Hill}}} \tag{3}$$

$c_{i,j}^{L}$ is the interaction coefficient between metabolites and bacteria of the lower phase where the level of metabolites is low, and $c_{i,j}^{U}$ is the interaction coefficient between metabolites and bacteria of the upper phase where the level of metabolites is high. $w_{i,j}$ is the point at which $\frac{M_i}{B_j}$ is halfway between $c_{i,j}^{L}$and $c_{i,j}^{U}$. This value represents the threshold that determines at what ratio the switching occurs. K_{Hill} is the Hill constant which determines the sharpness of the switch between metabolites being consumed and produced, and the sharpness refers to how quickly this switch occurs.

2.2 Incorporating data

The experimental data informing this research were originally collected for the study by Jansma et al, 2023 [30]. In that study, a synthetic microbial community of three species; *E. coli, Pseudomonas fluorescens* and *Streptococcus salivarius,* was

designed to resemble the upper small intestinal microbiota, allowing the effects of probiotic supplementation on community-level metabolism to be investigated. Three experimental conditions were tested: a control group with no probiotics and two treatment groups supplemented with either *Streptococcus thermophilus* or *Lactobacillus casei*. Concentrations of 31 metabolites were measured over time to capture how each bacterial treatment altered metabolic outputs compared to baseline. For the current study, only the control dataset is used. The primary purpose of incorporating these experimental data is to inform the categorisation of metabolites in the model as produced, consumed, or non-monotonic. This ensures that the simulation reflects a realistic distribution of metabolite types. In the small-scale simulations, this results in 7 consumed, 3 produced, and 2 non-monotonic metabolites interacting with 3 bacterial species. For the larger simulations, the same ratio is maintained, with 14 consumed, 6 produced, and 4 non-monotonic metabolites interacting with 10 bacterial species. This approach allows the model to capture biologically meaningful system composition while remaining a tractable framework for exploring microbial-metabolite dynamics.

2.3 Simulations

To investigate the fundamental dynamics of the gut microbial ecosystem, a basic simulation was implemented in R (version 4.4.2) using the *deSolve* package for solving ODEs. The purpose of these simulations is to investigate how different $c_{i,j}$ values portray the behaviour of the metabolites and bacteria. The initial model represents a closed system, defined by setting the metabolite influx term $s_i[t]$ to zero for all metabolites. The initial conditions for bacteria, produced metabolites, consumed metabolites and non-monotonic metabolites were drawn from Uniform(0.05, 0.2), Uniform(e^{-3}, e^{-2}), Uniform(0.5, 1) and Uniform(e^{-3}, e^{-2}), respectively. The interaction coefficients ($c_{i,j}$) for produced and consumed metabolites were drawn randomly from a uniform distribution with ranges of [-1,0] and [0,1] respectively. For non-monotonic interactions, the effective interaction coefficient $c_{i,j}$ is dynamically calculated at each time step using a Hill-type function. Unlike produced or consumed metabolites, where $c_{i,j}$ is constant, the Hill function expresses $c_{i,j}$ as a function of the metabolite-to-bacteria ratio. This formulation allows the strength of the interaction to vary dynamically over time, capturing non-monotonic behaviour such as metabolic switching and overflow metabolism. The values for $c_{i,j}^{L}$ and $c_{i,j}^{U}$ were also obtained

randomly from uniform distributions, in the ranges of [-1,0] and [1,0] respectively. The K_{Hill} value was set to 2, the $w_{i,j}$values drawn from a uniform distribution in a range of [0.1, 1]. The death rate, d_j was fixed at 0.1 and the metabolite conversion coefficient, c_0, of each bacterium was set to 0.1 [28]. The carrying capacity was experimentally approximated and scaled to 14 from the total number of bacterial cells (14 billion per mL) [28]. The system was simulated over a duration of 10 time units, and the resulting trajectories of all bacterial and metabolite populations were visualised to observe the key behaviours emerging from the defined interactions. This visualisation allows for the characterization of the system's evolution across three distinct phases. The initial phase $(t = 0)$ represents the starting concentrations before any interactions have occurred. This is followed by the transient phase, a dynamic period where metabolite and bacterial concentrations change rapidly as the system is in flux. Finally, the simulation reveals the asymptotic phase, which describes the long-term behaviour where the system reaches a steady state, and the concentrations of all populations no longer change significantly.

To explore the influence of environmental nutrient availability on the microbial ecosystem, the model was transitioned from a closed to an open system by incorporating a constant influx term, $s_i[t]$, for externally supplied metabolites. It is not necessary to include a separate degradation term in the metabolite equations because the bacteria consume the metabolites. The effective loss of metabolites is already captured through the interaction terms in the model. The selection of values for the different influx values was based on a principled analysis of the model's internal dynamics rather than arbitrarily. We determined the system's characteristic rate of metabolite consumption by considering the product of typical metabolite concentrations (M_i), bacterial abundances (B_j), and their interaction coefficients ($c_{i,j}$). This calculation provided a biologically relevant scale against which to set the influx. Based on this, we established three distinct environmental regimes: a nutrient-poor setting $(s_i[t] \approx 0.01)$, where influx is comparable to or less than the maximum consumption rate, leading to resource competition; a balanced setting $(s_i[t] \approx 0.05)$, where nutrients are sufficiently available to sustain complex community interactions; and a nutrient-rich setting $(s_i[t] \approx 0.1)$, where influx exceeds consumption, shifting the primary limitation on bacterial growth from metabolites to the system's carrying

capacity. Simulating the model across these different metabolite allows for a systematic investigation of how nutrient supply fundamentally shapes community structure and stability.

2.4 Stability

To assess the stability of the gut microbiome model, a linear stability analysis was performed by evaluating the eigenvalues of the Jacobian matrix, J, computed at the steady state of each system. The Jacobian captures the local sensitivity of the system to small perturbations in metabolite and bacterial levels and is mathematically expressed as $Jx = \lambda x$, where λ are the eigenvalues of J. The sign of the real parts of the eigenvalues determines the nature of the steady state: if all $\Re(\lambda_i) < 0$, perturbations decay over time and the system returns to equilibrium, indicating a stable steady state; if any $\Re(\lambda_i) > 0$, perturbations grow and the steady state is unstable; and if some $\Re(\lambda_i) \approx 0$, the system is marginally stable and exhibits slow relaxation dynamics. Eigenvalues with non-zero imaginary parts indicate oscillatory modes [31]. Equilibria for both closed systems (no metabolite inflow) and open systems (with inflow to consumed metabolites) were computed numerically using *multiroot* from the *rootSolve* package. The Jacobian was calculated numerically with *numDeriv::jacobian,* and eigenvalues were plotted on the complex plane to visualise stability. Metabolite and bacterial trajectories were also integrated over time using *deSolve::ode* to confirm that the qualitative dynamics agreed with the predictions from the linear stability analysis. This approach allowed quantitative comparison of open versus closed systems and provided insights into how interaction coefficients and system inputs influence the equilibrium behaviour of the gut microbiome. This was done in the *Simulations* section of *Methods*. In order to quantify the overall system stability, a single numerical measure was used and is known as the stability index. This is calculated by $S = max(\Re(\lambda_i))$, where $S < 0$ indicates a stable system, $S \approx 0$ indicates marginal stability, and $S > 0$ indicates instability. This is calculated for both the open and closed, small and large systems to be compared.

Bifurcation diagrams were also used in this study to assess the stability of the gut microbiome systems by examining how steady-state abundances of bacteria and metabolites change in response to variations in key interaction parameters. For

produced and consumed metabolites, the interaction coefficients $c_{i,j}$ are systematically varied across their respective ranges to evaluate how changes in these interactions influence system equilibria. In the case of non-monotonic metabolites, the Hill-switch parameter $w_{i,j}$ is varied to capture the effects of metabolic switching on stability. By plotting the resulting steady-state values as a function of these parameters, bifurcation diagrams provide a visual representation of the conditions under which the system maintains stable equilibria, transitions to alternative states, or exhibits potential instabilities. This approach enables a systematic exploration of the parameter space and identifies which interactions have the greatest influence on the robustness and resilience of the microbial community.

2.5 Networks

One of the main aims of carrying out network analyses is to determine keystone species in a community, which are taxa that play an important role within the community. These contributions can range from stability, productivity or richness in diversity [15]. The relationships between bacterial abundances and metabolite concentrations over time can be captured in the form of an association network, where nodes represent bacteria and metabolites, and edge weights represent the strength of their interactions. In this study, the weight of an edge between a metabolite M_i and a bacterium B_j is calculated as $M_i \times B_j \times c_{i,j}$, reflecting the influence of both the current abundance and the interaction coefficient. These networks are constructed at multiple time points to capture different phases of system dynamics: initial conditions, transient behaviour, and asymptotic steady states. Analyses are conducted for both a small network, which allows clear interpretation of fundamental interaction patterns, and a larger network, which demonstrates scalability and the emergence of more complex structures. When combined with host-microbe bipartite networks, such approaches have been used to reveal nested and hierarchical structures characteristic of healthy gut microbiota [28]. This network-based framework thus provides a complementary perspective to dynamical simulations, enabling the exploration of microbe-metabolite interactions across time and system size.

3 Results

The results are organised to address the core research questions systematically. First, baseline dynamics of the closed system are examined to establish fundamental metabolic and bacterial behaviours (Section 3.1). Second, the mechanisms underlying metabolic switching in non-monotonic metabolites are investigated to demonstrate how interaction coefficients vary dynamically with system state (Section 3.2). Third, the influence of environmental nutrient availability on system dynamics is explored by comparing closed and open systems across multiple influx scenarios (Section 3.3). Fourth, stability analysis via Jacobian eigenvalues quantifies the resilience of different system configurations to perturbations (Section 3.4). Finally, network analysis reveals the structural organization and temporal evolution of metabolite-bacteria interactions (Section 3.5).

3.1 Baseline Dynamics in Closed Systems

To establish a reference point for interpreting more complex scenarios, the model was first simulated under closed-system conditions where no external metabolite influx occurs ($s_i = 0$). The dynamics of the bacteria show that the levels increase initially due to the availability of metabolites, and then the levels (Figure 1A).

Consumed metabolites begin at relatively high concentrations and decline steadily as bacteria utilize them for growth, eventually plateauing as the system approaches equilibrium (Figure 1B). Produced metabolites start at low concentrations and show a gradual increase before stabilising (Figure 1C). Non-monotonic metabolites, which are governed by Hill-type switching functions, show intermediate behaviour that depends on the metabolite-to-bacteria ratio over time (Figure 1D).

The variability observed across different metabolites within each category arises from differences in their interaction coefficients ($c_{i,j}$), which determine the strength and direction of metabolite-bacteria interactions. These baseline dynamics demonstrate that in the absence of external influx, the system results in a stable steady state where metabolite concentrations and bacterial abundances no longer change.

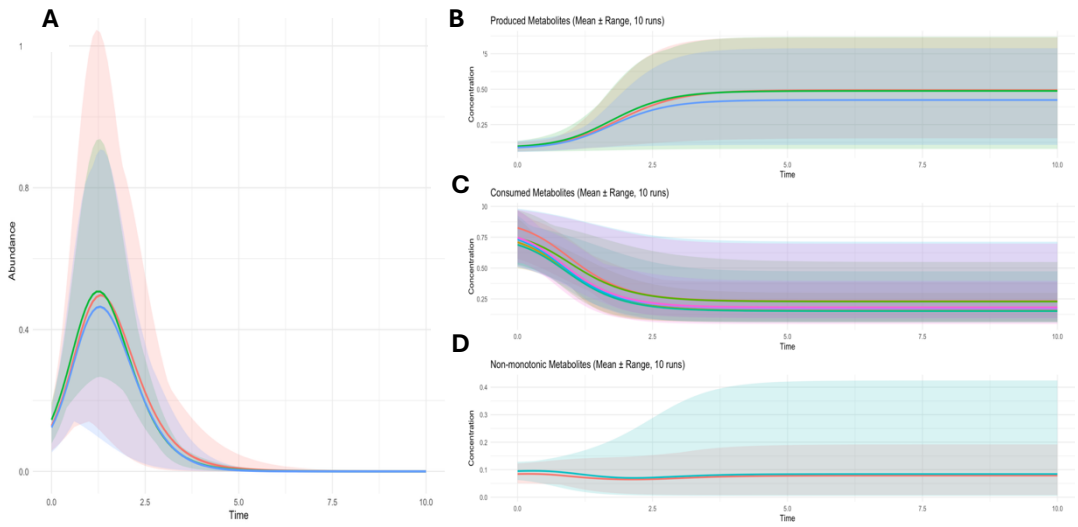

Figure 1: Baseline dynamics in a closed system.

Bacterial populations (A) and concentrations of produced (B), consumed (C), and non monotonic (D) metabolites. Each plot displays a simulation across 10 time units in a closed system with no external influx. Bacteria grow initially then decline as resources deplete. Produced metabolites accumulate, consumed metabolites decline, and non-monotonic metabolites show intermediate behaviour. Lines show mean values and ribbons show min-max range across 10 simulation replicates with different random parameters.

3.2 Demonstration of Metabolic Switching

A crucial feature of the model is its ability to capture metabolic switching, where the nature of a metabolite's interaction with bacteria changes depending on the relative abundances of metabolites and bacteria in the system. The Hill function (Eq. 3) describes the relationship between the interaction coefficient $c_{i,j}$ and the metabolite-to-bacteria ratio ($\frac{M_i}{B_j}$) where the Hill coefficient k_{Hill}, which controls the steepness of the transition. When the $\frac{M}{B}$ ratio is low the interaction coefficient approaches its upper limit $c_{i,j}^{U} = 1$, indicating that bacteria consume the metabolite and benefit from it for growth. As the $\frac{M}{B}$ ratio increases beyond a threshold value the interaction coefficient transitions through zero and approaches its lower limit $c_{i,j}^{L} = -1$, indicating that bacteria switch to producing the metabolite (Figure 2). The sharpness of this transition depends critically on k_{Hill}, where lower values produce a gradual, smooth transition, while higher values produce an abrupt, switch-like behaviour. The boundary between consumption and production can be represented by $\mathrm{c_{i,j}} = 0$. This switching behaviour

reflects biologically realistic phenomena such as overflow metabolism and concentration-dependent toxicity, where metabolites like formate or acetate can support growth at low concentrations but become inhibitory when they accumulate to high levels. The Hill coefficient $k_{Hill} = 2$ used in the simulations throughout this study (as specified in Equation 3) represents a relatively gentle transition, allowing the system to exhibit gradual metabolic shifts rather than instantaneous switches.

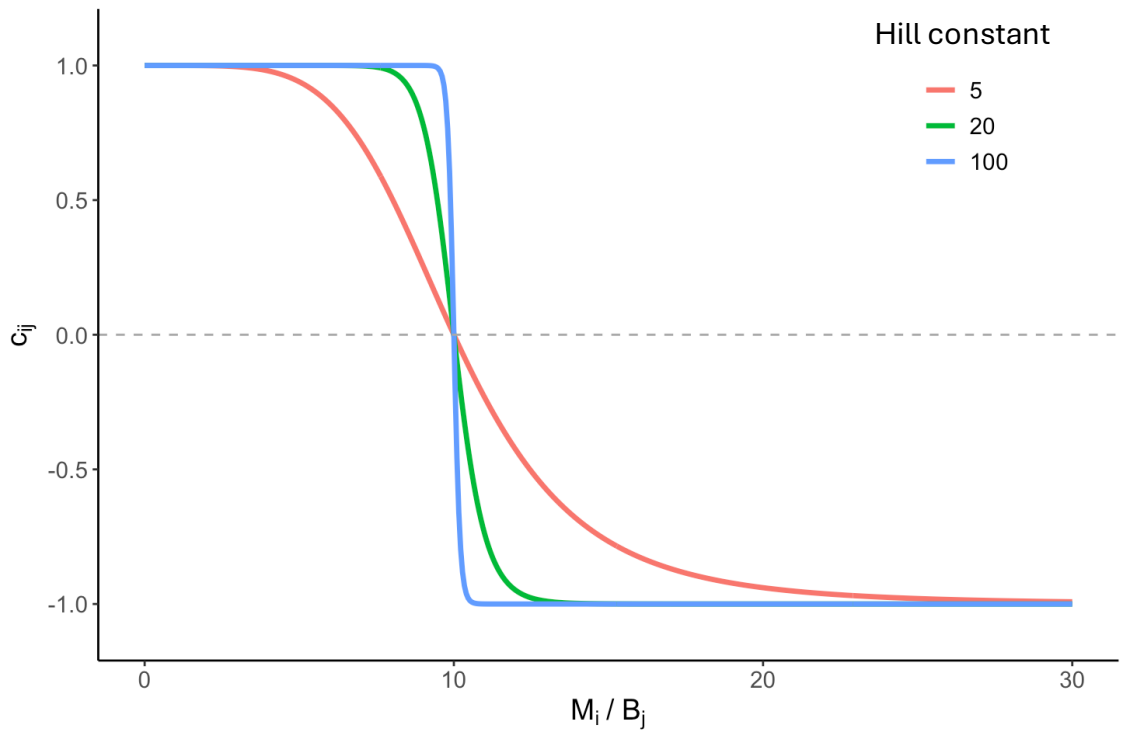

Figure 2 Theoretical Hill function for metabolic switching.

The interaction coefficient $c_{i,j}$ varies with the metabolite-to-bacteria ratio (M_i/B_j). Three curves show different Hill coefficient values (k = 5, 20, 100). The dashed line indicates the transition between production ($c_{i,j}$> 0) to production ($c_{i,j}$ < 0).

3.3 Influence of Nutrient Influx on System Dynamics

To investigate how environmental nutrient availability shapes microbial-metabolite dynamics, the model was used to incorporate constant external influx of consumed metabolites at three different rates: nutrient-poor (s_i ≈ 0.01), balanced (s_i ≈ 0.05), and nutrient-rich (s_i ≈ 0.1). Each scenario was simulated 10 times with different randomly generated parameter sets, and the mean trajectories are shown with ribbons indicating the minimum-maximum range across replicates (Figure 3).

Bacterial dynamics reveal a consistent pattern across all three influx scenarios: populations initially grow as they consume the available metabolites, reaching peak abundances between $t = 0.5$ and $t = 5$, before declining toward extinction by $t = 10$ (Figure 3A-C). The wide ribbons during the growth phase indicate substantial variability in peak bacterial abundances across simulation replicates, reflecting sensitivity to randomly generated interaction coefficients.

Consumed metabolites show rapid depletion from their initial high concentrations as bacteria utilise them during the early growth phase. After the initial decrease, these metabolites stabilize at equilibrium concentrations that are proportional to the influx rate: nutrient-poor systems maintain lower steady-state levels (~0.5-0.6) (Figure 3D), while nutrient-rich systems maintain higher levels (~0.6-0.7)(Figure 3F). The balanced system also shows a more rapid increase (Figure 3E). However, these sustained metabolite levels are insufficient to prevent bacterial extinction, indicating a fundamental imbalance in the system dynamics (Figure 3A).

Non-monotonic metabolites exhibit steady accumulation over time, with the rate of increase directly proportional to the nutrient influx. In the nutrient-poor scenario, these metabolites accumulate slowly and plateau around 0.3-0.4 (Figure 3D), whereas in the nutrient-rich scenario (Figure 3F), they accumulate rapidly, exceeding concentrations of 2.0 by $t = 20$. The balanced system also shows a more rapid increase (Figure 3E). This linear accumulation pattern suggests that these metabolites are primarily being produced rather than consumed, and the Hill switching mechanism is not being triggered due to the declining bacterial populations.

Produced metabolites show similar accumulation patterns to the non-monotonic metabolites, increasing steadily over time with rates proportional to influx. These metabolites are generated as byproducts of bacterial metabolism during the early growth phase and continue to accumulate even after bacteria decline. The variability across replicates (indicated by ribbon width) is relatively narrow for these metabolites, suggesting that their production is largely deterministic once bacterial growth occurs.

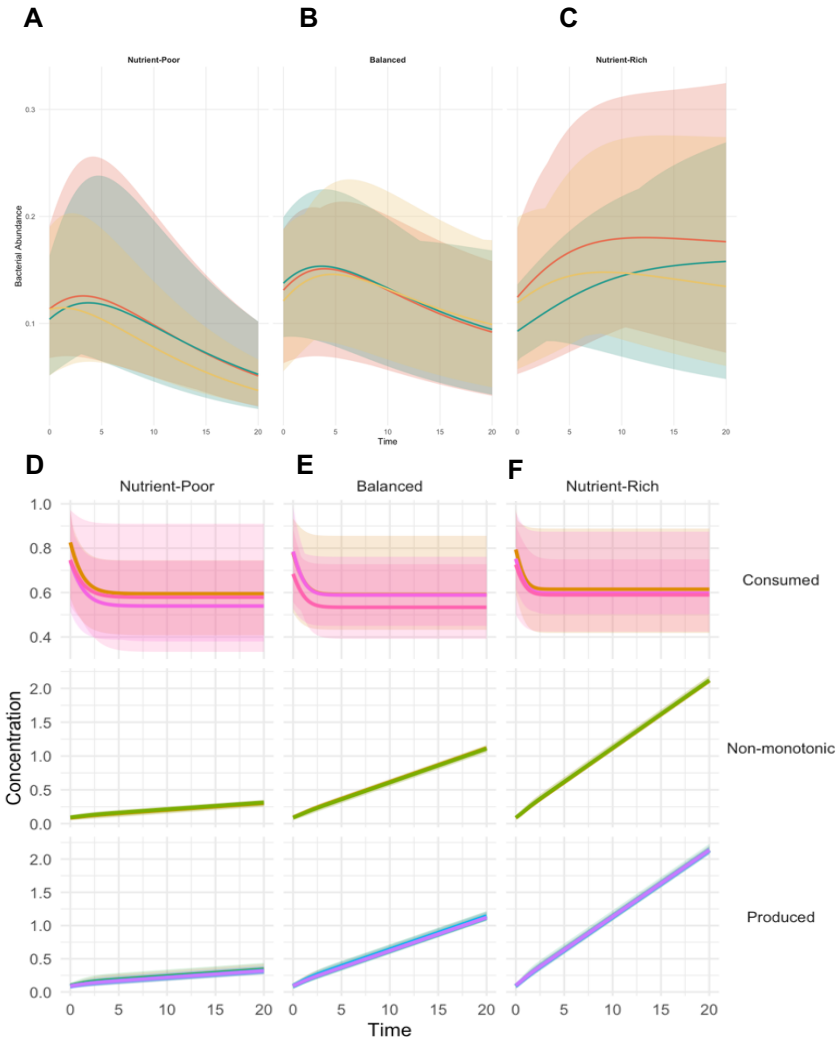

Figure 3: Bacterial and metabolite dynamics across three nutrient influx scenarios.

Top panels show bacterial abundances for three species (B1, B2, B3) in nutrient-poor (s_i= 0.01), balanced (s_i = 0.05), and nutrient-rich (s_i = 0.1) open systems. Bacteria initially grow but decline toward extinction in all scenarios by t = 10. Lower panels show metabolite dynamics grouped by type: consumed metabolites (top) deplete initially then stabilise at levels proportional to influx; non-monotonic metabolites (middle) and produced metabolites (bottom) accumulate linearly over time, with accumulation rates increasing with nutrient availability. Lines represent mean trajectories and ribbons show min-max range across 10 simulation replicates.

3.4 Stability Analysis

To assess the stability and resilience of the gut microbiome model, linear stability analysis was performed by computing the Jacobian matrix at equilibrium points and analysing its eigenvalues. The stability of a system is determined by the sign of the real parts of its eigenvalues: negative real parts indicate that perturbations decay over time (stable equilibrium), positive real parts indicate perturbations grow (unstable equilibrium), and real parts near zero indicate marginal stability with slow relaxation dynamics. Furthermore, bifurcation diagrams were also constructed in order to assess the effects of the range of certain parameters such as the interaction coefficient ($c_{i,j}$) as well as the Hill switch point ($w_{i,j}$).

3.4.1 Eigenvalue spectra and stability classification

Figure 4 presents the eigenvalue spectra for the small system in both closed and open configurations, with eigenvalues plotted in the complex plane. Figure 5 shows the corresponding eigenvalue spectra for the large system in closed and open configurations. Table 1 summarizes the stability index (S = max($\Re$(λ))) and stability classification for all four configurations based on 10 simulation replicates. The small system exhibits eigenvalues with positive real parts for the closed configuration (Table 1), indicating dynamic instability (Figure 4). The small open system shows reduced but still positive stability indices ($S = 0.27$), indicating persistent instability despite continuous metabolite influx (Figure 4, Table 1)). The large system displays eigenvalues with real parts very close to zero ($S = -0.008$) for the closed configuration, indicating marginal stability (Figure 5A, Table 1). The large open system shows positive stability indices ($S = 0.31$), with several eigenvalues in the right half-plane. Eigenvalues with non-zero imaginary parts are present in several configurations, indicating oscillatory dynamics (Figure 5B, table 1).

3.4.2 Bifurcation analysis

To investigate the relationship between bacterial abundances and the interaction parameters a bifurcation analysis was performed. For the consumed and produced metabolites, the interaction coefficient ($c_{i,j}$) was varied across its full range, [0, 1] and [-1, 0] respectively; for non-monotonic metabolites, the Hill switch point ($w_{i,j}$) was varied, as well as the upper and lower $c_{i,j}$ values. The upper values are ranged from [0, 1] and the lower, [-1, 0]. The representative example of two consumed metabolite-

bacteria pairs shows relatively smooth changes in bacterial abundances across the parameter range, with B3 maintaining moderate abundances (~0.06-0.12)and B1 showing peaks at both low and high $c_{i,j}$ values (Figure 6A). The M4-B2 pair exhibits a sharp bifurcation at $c_{i,j}$ ≈ 0.85, where B2 abundance suddenly increases from near-zero to approximately 0.5 while B1 and B3 decline to near-zero. Below this threshold, all three bacterial species coexist at low to moderate abundances; above it, B2 dominates (Figure 6B). The non-monotonic metabolites examples display stable equilibria across the entire range of Hill switch points ($w_{i,j}$ from 0.1 to 1.0). Bacterial abundances remain essentially constant regardless of the switching threshold, with B1 and B3 coexisting at steady levels (~0.075 each) while B2 remains near zero (Figure 6C & D). The representative examples of the produced metabolites show smooth, gradual changes in bacterial composition as $c_{i,j}$ varies from -1 to 0. For M1-B1, B1 declines gradually from ~0.09 to ~0.06 as production strength increases, while B3 remains relatively stable (Figure 6E). The M1-B2 pair exhibits a smooth trade-off pattern where B1 decreases and B3 increases as production strengthens (Figure 6F). No sharp bifurcations are observed for produced metabolites.

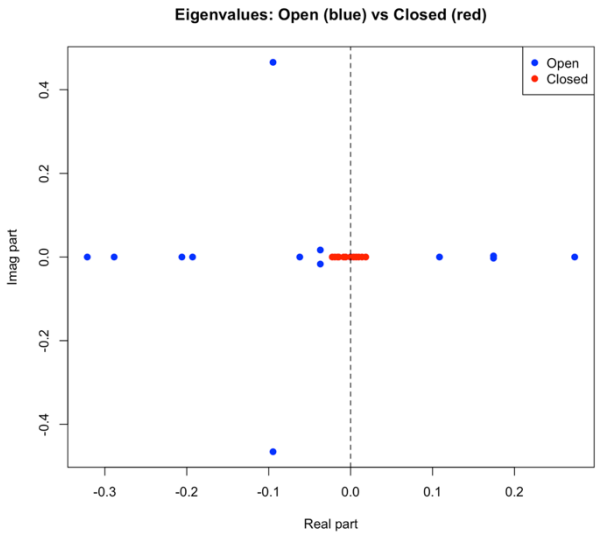

Figure 4: A comparative plot of the eigenvalue spectra for a small open system and closed system.

The eigenvalues for the closed system all have non-positive real parts, clustering around the imaginary axis. In contrast, the open system displays several eigenvalues with positive real parts. The grey dots indicate the axis separating the positive and negative values of the real parts of the eigenvalues.

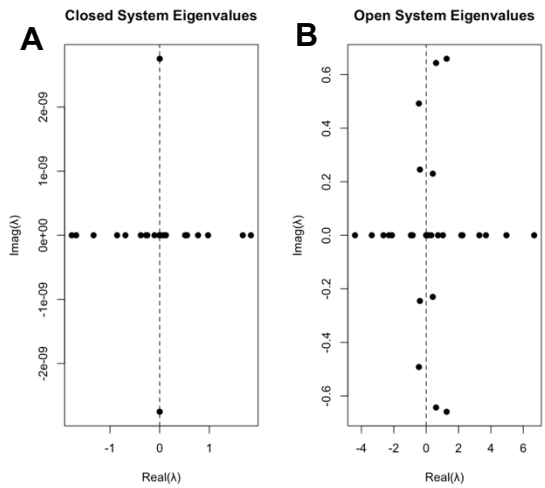

Figure 5: Eigenvalue spectra for a large closed system (A) and open system (B).

The closed system shows eigenvalues with non-positive real parts, clustering near the imaginary axis. The y-axis scale is on the order of 10^{-9}, indicating that the imaginary parts of all eigenvalues are effectively zero, meaning the system approaches its steady state without oscillations. (B) The open system displays multiple eigenvalues with positive real parts (right half of the complex plane). The presence of significant non-zero imaginary parts, appearing as complex conjugate pairs, reveals oscillatory dynamics in the system.

***Table 1:* Summary of different system sizes and types and their stability index based on the maximum real part of the eigenvalues in the Jacobian**

System size	System type	Stability index ($max(\Re(\lambda))$)	Stability classification
Small	Closed	2.681456	unstable
	Open	0.2735005	unstable
Large	Closed	-0.007659672	Marginal stable
	Open	0.3134814	unstable

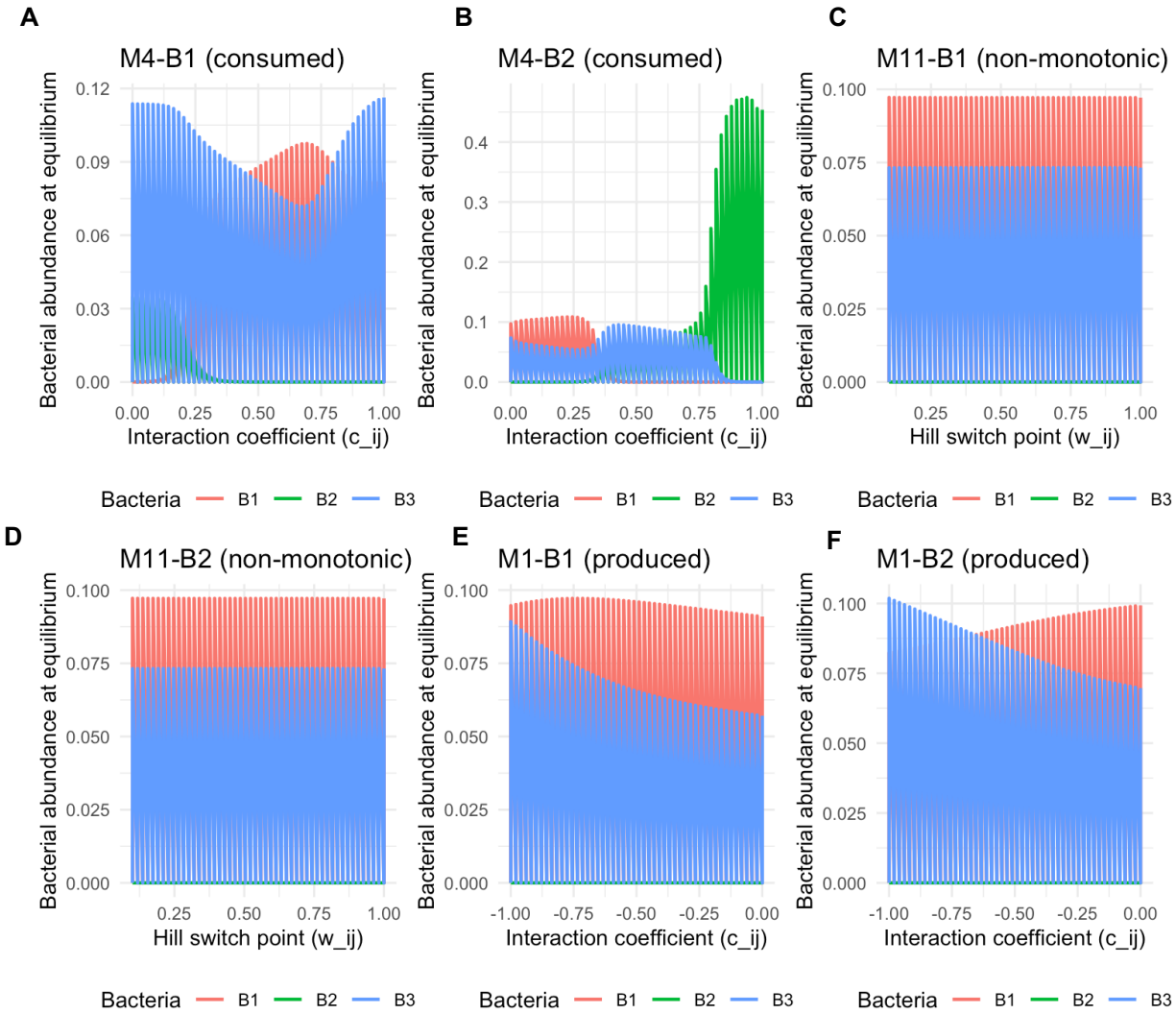

Figure 6: Bifurcation diagrams of bacterial equilibria across interaction parameter ranges.

Stacked areas show abundances of B1 (red), B2 (green), and B3 (blue) at equilibrium as interaction coefficients ($c_{i,j}$) or Hill switch points ($w_{i,j}$) are varied for consumed (M4), non-monotonic (M11), and produced (M1) metabolite types. M4-B2 shows a critical bifurcation at $c_{i,j} \approx 0.85$ where B2 suddenly dominates. Non-monotonic metabolites maintain stable equilibria across all switching thresholds. Produced metabolites show smooth compositional changes with no sharp transitions. Vertical striations are plotting artifacts from discrete sampling.

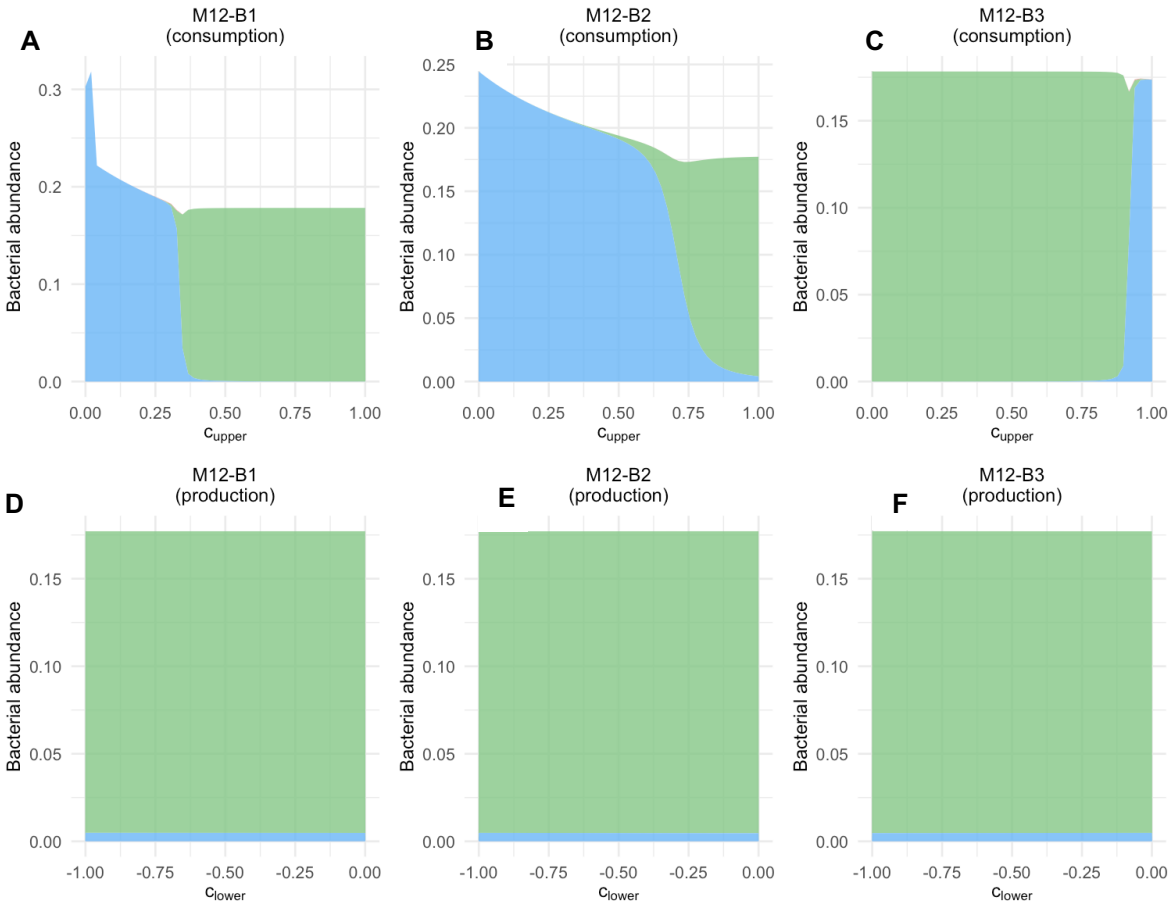

Figure 7 Bifurcation analysis of non-monotonic metabolite M12 varying consumption and production strengths.

Top row: varying $c_{i,j}^{U}$ (0 to 1) shows smooth transitions from B3 to B2 dominance as consumption strength increases. Bottom row: varying $c_{i,j}^{L}$ (-1 to 0) reveals minimal equilibrium changes, with B2 maintaining stable dominance. The asymmetric responses reflect that only consumption (positive $c_{i,j}$) contributes to bacterial growth in the model, while production (negative $c_{i,j}$) does not directly affect growth. Stacked areas show abundances of B1 (red), B2 (green), and B3 (blue).

3.5 Network Structure and Temporal Evaluation

To explore the structural organisation of metabolite-bacteria interactions, network analysis was conducted at three distinct time points: initial ($t = 0$), transient ($t = 2$), and asymptotic ($t = 20$). Networks were constructed by representing metabolites and bacteria as nodes, with directed edges weighted by the product $M_i \times B_j \times c_{i,j}$, reflecting the combined influence of abundance and interaction strength. The evolution of the network structure of a small system under closed-system conditions from at the initial time point ($t = 0$) (Figure 8A) to the transient phase (Figure 8B) shows a transition from are relatively sparse relatively sparse, dominated by edges from the consumed metabolites to bacteria to a more dense and more complex network. During this transition, the consumed metabolites begin at high concentrations while produced and non-monotonic metabolites are scarce. During the transient phase ($t = 2$), the network bacterial populations grow, and metabolite concentrations adjust. Consumed metabolite edges weaken as these resources are depleted, while produced metabolite edges strengthen as bacteria generate byproducts. Non-monotonic metabolites exhibit dynamic switching, with edges transitioning between beneficial and inhibitory interactions depending on the evolving $\frac{\mathrm{M_i}}{\mathrm{B_j}}$ ratios. During the asymptotic phase ($t = 20$) (Supplementary figure 5), the network becomes sparse again as bacterial populations decline toward extinction in the closed system, causing all edge weights to approach zero since they depend on bacterial abundance.

In the large system, similar temporal patterns emerge but with greater structural complexity due to the increased number of metabolites and bacteria (Figure 10A and B, supplementary figure 5.) The disappearance of edges at $t = 20$ highlights a key limitation of closed systems: without external nutrient input, bacterial populations cannot be sustained indefinitely, leading to ecosystem collapse. These network visualisations reveal how interaction strength and directionality are emergent properties that depend critically on system state, rather than being static features of the microbial community.

Small network

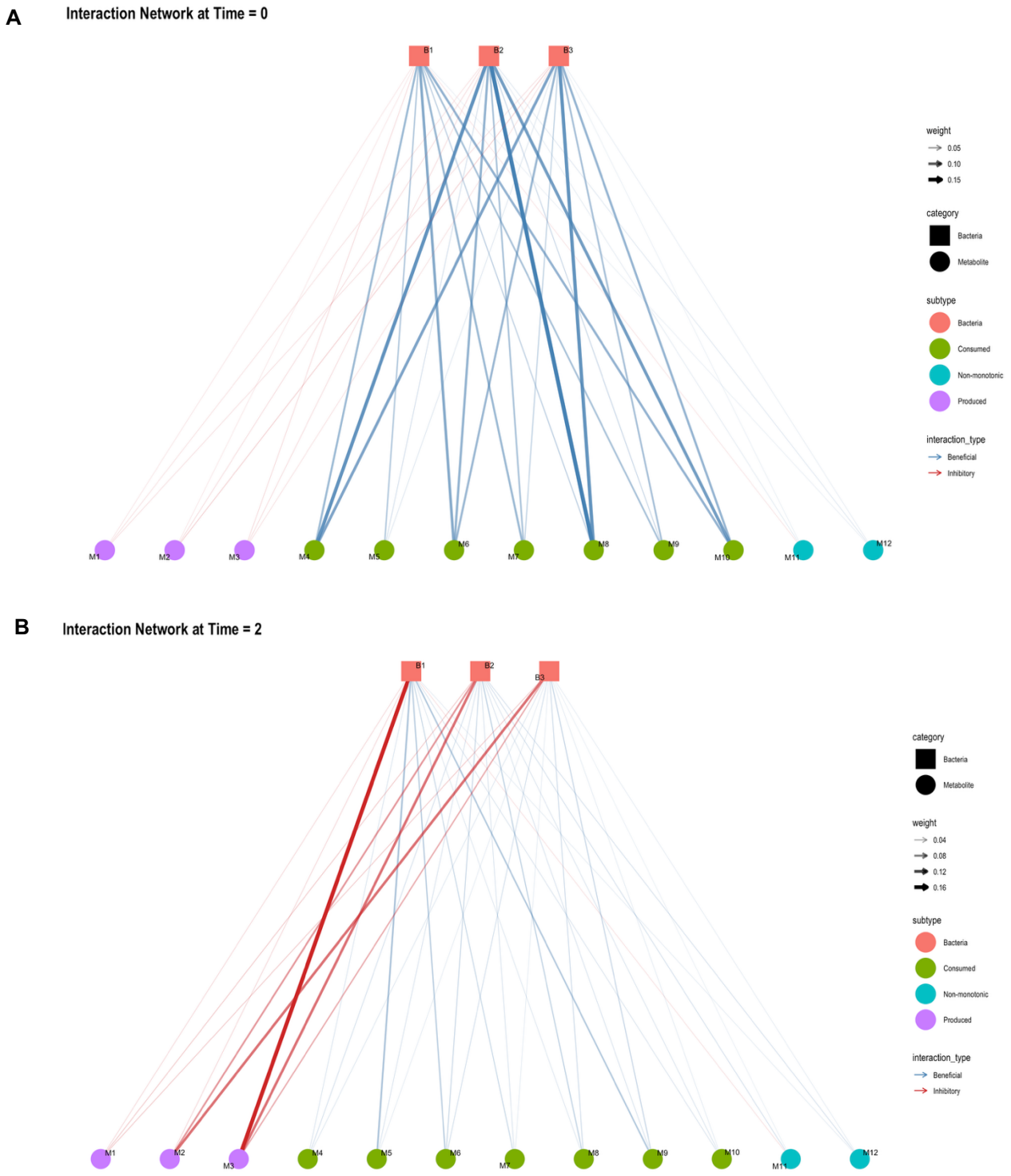

Figure 8: Metabolite-bacteria interaction networks at initial (t = 0) and transient (t = 2) time points in a closed system. *Bacteria are shown as squares (B1-B3) and metabolites as circles (M1-M12), coloured by type: produced (purple), consumed (green), non-monotonic (purple). Edge width represents interaction strength (M_i × B_j × c_ij); edge colour indicates interaction type (blue = consumption, red = production). At t = 0, consumption interactions dominate from high-abundance consumed metabolites. By t = 2, inhibitory interactions strengthen as bacteria grow and produce metabolites, while consumption edges weaken as resources deplete. Non-monotonic metabolites exhibit both edge types simultaneously, demonstrating metabolic switching.*

Large network

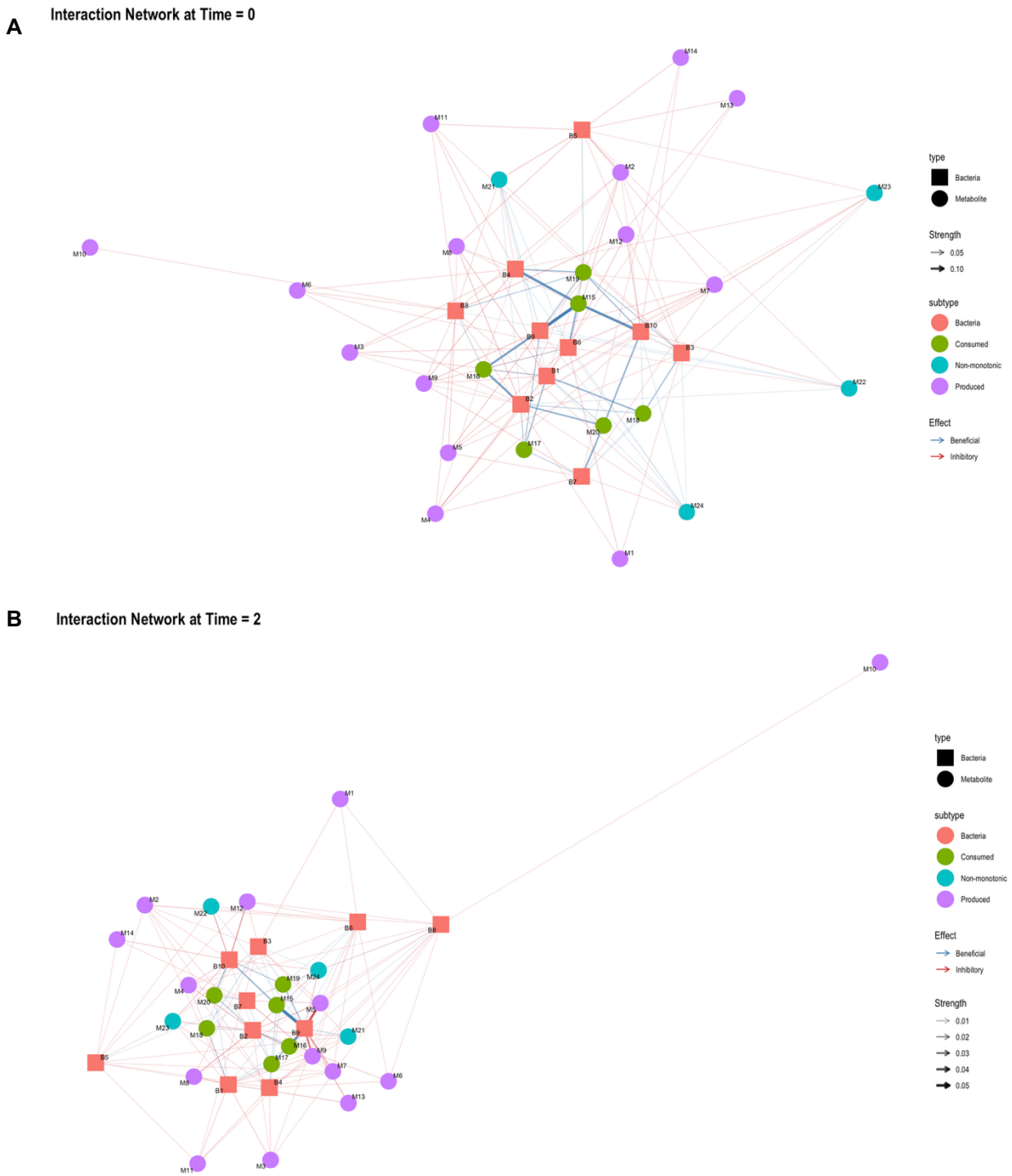

Figure 9: ***Metabolite-bacteria interaction networks for the large system (24 metabolites, 10 bacteria) at t = 0 and t = 2.*** *Bacteria (squares, B1-B10) and metabolites (circles, M1-M24) are coloured by type: produced (purple), consumed (green), non-monotonic (purple). Edge width indicates interaction strength (M_i × B_j × c_ij); edge colour shows interaction type (gray = beneficial, red = inhibitory). At t = 0, the network is highly connected with predominantly consumption interactions. By t = 2, the network reorganizes into a more modular structure with stronger, more selective interactions and some peripheral nodes becoming isolated.*

Table 2: summary of network properties of the small and open systems

Network property	*Small system*	*Large system*
Number of nodes	15	34
Number of edges	36	129
Network density	0.1714	0.1150
Average degree	4.80	7.59
Global clustering coefficient	0	0
Modularity score	0.1960	0.3361
Number of communities	3	4
Nestedness score	NA	NA

4 Discussion

This study aimed to mathematically model the dynamics of gut microbial-metabolite interactions to understand how metabolic switching and external nutrient inputs influence community stability. We hypothesized that open systems, characterised by continuous metabolite influx, would exhibit greater instability than closed systems. The results partially contradicted this hypothesis: while small systems were unstable regardless of influx, large closed systems achieved marginal stability ($S \approx 0$), and large open systems remained unstable ($S > 0$). Dynamic simulations revealed that closed systems converged to steady states where bacteria went extinct, whereas open systems displayed persistent oscillatory dynamics with boom-bust bacterial cycles. Stability analysis confirmed these observations through eigenvalue spectra, with positive stability indices indicating instability and non-zero imaginary parts reflecting oscillatory behaviour. Bifurcation analysis showed that bacteria can coexist across most parameter values, with only a few critical points causing sudden shifts where one species takes over and excludes the others. Network analysis showed temporal evolution from dense, consumption-dominated structures to sparse configurations as bacteria declined. These findings reveal that system size, interaction type, and external influxes together determine stability, with important implications for understanding microbiome resilience and designing therapeutic interventions.

4.1 Key findings

4.1.1 Simulations

The baseline simulations of the closed small system produce relatively simple dynamics, but they provide an important point of reference for interpreting more complex scenarios. Consumed metabolites begin at high concentrations and steadily decline as they are used by the bacteria, eventually stabilising once the system reaches equilibrium. Produced metabolites, in contrast, start at lower concentrations and increase over time before plateauing at a constant concentration. The plateauing behaviour in both cases reflects the fact that, in the absence of external inputs or outputs, the system eventually reaches a steady state where metabolite levels no longer change. This is because in a closed system, bacteria consume available metabolites until they reach carrying capacity. At this point, bacterial growth stops, so metabolite consumption/production also stops. The system reaches equilibrium where the rates of metabolite consumption and production balance out, causing

concentrations to plateau. Variability between different metabolites of the same type arises from the differences in their interaction coefficients (c_{ij}), which capture how strongly each metabolite interacts with the bacterial populations. These differences determine the rate and extent of metabolite consumption or production. While the overall patterns appear trivial, they highlight the role of c_{ij} in shaping system behaviour and demonstrate how the system tends to settle into stable metabolite levels in a closed environment. This baseline provides a useful comparison for later simulations with additional dynamics, such as non-monotonic interactions and metabolic switching. Both the small and large system exhibited the same characteristic dynamics: consumed metabolites declined from higher initial concentrations before plateauing, while produced metabolites decreased from lower initial levels until they reached a steady state.

One notable observation is that the non-monotonic metabolites do not exhibit visible switching behaviour in the time-series plots (Figure 1), despite being governed by Hill-type functions designed to capture metabolic switching. This absence of switching does not indicate a failure of the model but rather reflects the specific conditions under which switching occurs. Metabolic switching, as defined by Equation 3, depends on the metabolite-to-bacteria ratio ($\frac{M_i}{B_j}$) crossing the threshold parameter $w_{i,j}$. In the closed system bacteria grow rapidly during the initial phase (t = 0 to t = 2) and then decline toward extinction as resources are depleted. During this trajectory, the $\frac{M_i}{B_j}$ ratio for non-monotonic metabolites may never cross the critical threshold that would trigger a transition from one metabolic mode to another. For switching to be visible in the time-series, the system would need to progress through a phase where $\frac{M_i}{B_j}$ is low (bacteria consuming the metabolite, $c_{i,j}$> 0), followed by a phase where $\frac{M_i}{B_j}$ increases substantially (bacteria producing the metabolite, $c_{i,j}$ < 0), creating a clear inflection point in the metabolite trajectory. However, in the closed system, bacteria decline so rapidly that non-monotonic metabolites remain at low concentrations throughout, keeping the $\frac{M_i}{B_j}$ratio predominantly on one side of the switching threshold. Additionally, the particular values of $w_{i,j}$ (drawn randomly from the range 0.1 to 1.0) and the initial conditions may position the system in a parameter regime where switching is

theoretically possible but not realized during the simulated timeframe. This phenomenon is comparable to a bifurcation diagram showing that alternative stable states exist at certain parameter values, but a particular simulation starting from specific initial conditions may never visit those regions of state space. To observe switching behaviour in simulations, it would be necessary to engineer scenarios where bacteria first grow substantially (driving M/B low and promoting consumption), then decline while metabolites continue to accumulate from external sources or slow degradation (driving $\frac{M_i}{B_j}$high and triggering production). This could potentially occur in open systems with carefully tuned influx rates, or in multi-phase dynamics where bacterial populations undergo multiple boom-bust cycles that sweep the M/B ratio across the threshold. Future work could systematically explore initial conditions and parameter combinations that maximize the likelihood of observable switching, or implement time-varying influx rates that deliberately drive the system through the switching threshold to demonstrate this regulatory mechanism in action [12].

4.1.2 Stability

Stability and resilience are key ecological characteristics of the gut microbiome and are essential for assessing host health. Factors such as microbial diversity, metabolic flexibility, and both microbe-microbe and host-microbe interactions influence the system's resilience. Perturbations can disrupt the equilibrium of the gut microbiome, leading to reductions in functionality and diversity, which ultimately compromise host health [32].

The stability analysis of the small and large systems reveals important insights into how system size and external inputs influence microbial-metabolite dynamics. The stability indexes are summarised in Table 1. In the small system, both the closed ($S = 2.68$) and open ($S = 0.27$) systems are unstable, with perturbations growing rapidly, although the open system is slightly less unstable. In contrast, the large system exhibits a marginally stable closed system ($S = -0.008$), indicating slow relaxation to equilibrium, while the open system ($S = 0.31$) becomes unstable, albeit with slower perturbation growth compared to the small system. These results suggest that larger microbial-metabolite networks are inherently more resilient when isolated from

external metabolite inputs, but the addition of external substrates can destabilise the system.

These results align with the ecological theory linking diversity to stability, as demonstrated by *Lozupane et al*. (2012.). The authors demonstrated that higher microbial diversity is associated with an increase in resilience in the human gut microbiome, and that diverse communities are better able to resist perturbations and recover from disturbance [9]. The faster growth of perturbations in small systems highlights their sensitivity to fluctuations, reflecting the limited buffering capacity of simple microbial communities. Furthermore, the marginal stability that is achieved in the large closed system is consistent with the findings of *Coyte et al*. (2015.) Here, the authors found that complex microbial networks can achieve stability when metabolic competition is balanced by cross-feeding interactions [31]. The faster growth of perturbations in small systems aligns with how they limited buffering capacity, which is consistent with communities with low diversity that receive antibiotic treatment are more susceptible to the invasion of pathogens and recover slower to homeostasis [6]. Overall, these findings emphasise that the diversity of a system and the influxes into the environment together shape stability, with larger networks achieving resilience through the interactions of species within the environment, as opposed to the external inputs alone.

The observed differences in stability between small and large systems have clear biological implications for host health. A stable gut microbiome is generally associated with robust metabolic function, resistance to pathogen invasion, and maintenance of immune homeostasis [32]. In the large closed system, the marginally stable dynamics suggest that the microbial community can buffer small perturbations, helping maintain equilibrium and consistent metabolite levels, which is beneficial for the host. Conversely, the instability observed in open systems, and in smaller networks, indicates that fluctuations in metabolite concentrations or microbial abundances could grow over time, potentially leading to dysbiosis. Such instability could impair the gut's metabolic output, reduce diversity, and increase susceptibility to disease. These results highlight that both the size and connectivity of microbial-metabolite networks, as well as exposure to external inputs, may influence the resilience of the microbiome,

with direct implications for maintaining host health and preventing chronic gut-related disorders [33].

The bifurcation analysis revealed key differences in how interaction strength affects community stability across metabolite types. Consumed metabolites exhibited sharp bifurcations, exemplified by the M4-B2 pair where a critical threshold at $c_{i,j} \approx 0.85$ triggered sudden competitive exclusion, with B2 abundance jumping from near-zero to 0.5 while displacing B1 and B3 (Figure 6). This discontinuous response demonstrates the existence of alternative stable states and highlights the fragility of communities structured by resource competition, where small changes in interaction strength can precipitate regime shifts reminiscent of dysbiotic transitions observed clinically [12]. In contrast, non-monotonic metabolites displayed smooth transitions: varying consumption strength ($c_{i,j}^{U}$) for M12 produced gradual shifts from B3 to B2 dominance without any critical thresholds, while varying production strength ($c_{i,j}^{L}$) yielded essentially flat responses with stable B2 dominance across all parameter values (Figure 7). The insensitivity to production strength reflects the model structure where only consumption ($c_{i,j}^{U}$) contributes to bacterial growth (Equation 2). Produced metabolites showed intermediate behaviour with smooth, gradual compositional changes as production strength varied, indicating predictable rather than catastrophic responses. These contrasting patterns suggest that therapeutic interventions targeting non-monotonic or produced metabolites would yield controllable, proportional effects, whereas manipulating consumed metabolites carries risk of triggering sudden community collapse [18], [31]. The robustness of non-monotonic interactions across both switching thresholds ($w_{i,j}$, Figure 6) and interaction magnitudes ($c_{i,j}^{U}$) demonstrates that metabolic flexibility provides dual benefits: adaptive feedback through switching and stable compositional responses to parameter variation, supporting the hypothesis that metabolically versatile bacteria enhance community resilience [12].

4.1.3 Networks

In constructing larger microbial-metabolite networks, it is biologically relevant to set many interaction coefficients to zero, reflecting the fact that not all metabolites are utilised or produced by every microbial species. In real microbial communities, species exhibit substrate specificity and metabolic specialisation [12], meaning that only a

subset of metabolites directly influences the growth or activity of each bacterium. Representing this in the model enhances biological realism and prevents the artificial assumption that every metabolite affects every species equally. For example, in a small community consisting of *Lactobacillus plantarum* and *Escherichia coli*, *L. plantarum* produces lactate but does not interact with formate, while *E. coli* consumes formate and produces acetate. In this case, the interaction coefficients between *L. plantarum* and formate, and between *E. coli* and lactate, are set to zero, whereas the coefficients representing lactate's effect on *E. coli* and formate's effect on *L. plantarum* are maintained according to known metabolic roles [34]. This selective connectivity captures the structured and specialized nature of microbial–metabolite interactions, ensuring that the network reflects realistic patterns of dependency and influence within the system. Environmental perturbations can shift species abundance patterns and alter interaction network [35].

At different time points, the structure of the metabolite-bacteria interaction network reveals how the system evolves from a highly connected, dynamic state toward a stable, specialised configuration. At the initial stage ($t = 0$), both the metabolite-metabolite ($WW^\top$) and bacteria–bacteria ($W^\top W$) networks are dense, reflecting widespread interactions as the system adjusts to initial conditions and metabolites and bacteria interact broadly without clear organization. By the intermediate stage ($t = 2$), distinct clusters begin to emerge, indicating the onset of functional specialization—certain bacteria preferentially associate with metabolites, leading to the formation of metabolic modules or niches. At equilibrium ($t = 20$), most interactions weaken or disappear, resulting in a sparse network structure. This sparsity reflects niche stabilisation, where bacteria occupy specific roles with limited overlap, and metabolite fluxes have reached steady state. Together, these temporal shifts illustrate how dynamic microbial-metabolite networks self-organize over time, transitioning from generalist to specialist configurations as the system moves toward metabolic equilibrium [36], [37].

The network analysis highlights clear structural differences between the small and large systems, reflecting how increasing community size and complexity affect ecological connectivity [15]. The larger system contained more nodes (34 vs. 15) and edges (129 vs. 36), indicating a richer and more interconnected interaction network.

Despite this, its network density decreased (0.1150 compared to 0.1714), suggesting that as the system expands, individual connections become more sparse. The average degree, however, increased from 4.8 to 7.6, showing that on average, each node in the larger system interacts with more partners, potentially increasing metabolic interdependence. Both systems showed a global clustering coefficient of 0, meaning there were no tightly clustered subgroups where nodes are directly connected to one another, which is consistent with a bipartite network structure between metabolites and bacteria.

Both systems exhibited a global clustering coefficient of 0, which is expected for bipartite networks where metabolites connect only to bacteria and bacteria connect only to metabolites. This structure precludes the formation of triangular motifs required for clustering, as metabolites cannot directly interact with other metabolites, and bacteria cannot directly interact with other bacteria. All interactions are mediated across the two node types. This reflects the biological reality that microbial interactions occur indirectly through metabolic exchanges rather than direct contact.

Modularity, which is a measure of how well network can be divided into distinct groups, increased from 0.1960 in the small system to 0.3361 in the large system, and the number of communities rose from three to four. This implies that the larger system exhibits stronger compartmentalisation, with more distinct functionality. The relatively small increase in community number, despite the large system containing over three times as many bacterial species, indicates functional redundancy where multiple species occupy similar metabolic niches and cluster together into guilds. This contrasts with the small system, where each of the 3 bacteria essentially forms its own isolated community. The emergence of shared communities in the larger system reflects metabolic cooperation, where bacteria group based on functional relationships, for example cross-feeding, rather than operating independently. This pattern of functional modularity, where the number of communities is much smaller than the number of species, is consistent with observations from real gut microbiome studies. In the human gut, which has hundreds to thousands of bacterial species, microbial communities functionally organise into a limited number of major metabolic guilds or modules, typically ranging from 5 to 10 distinct functional groups [35]. These modules correspond to key metabolic processes such as primary degradation of

complex polysaccharides, production of short-chain fatty acids like butyrate and acetate that serve as cross-feeding substrates, metabolism of amino acids from protein breakdown, and synthesis of essential vitamins such as B vitamins and vitamin K [7]. This functional redundancy, where multiple taxonomically distinct species contribute to the same metabolic function, provides resilience to the microbial community: if one species is lost due to a perturbation, others within the same functional module can compensate and maintain overall metabolic output [32]. The modest number of communities detected in the current model (3 for the small system, 4 for the large system) reflects this principle of functional organisation, where metabolic role rather than species identity determines community structure.

Nestedness is a description of a hierarchical pattern where nodes with fewer connections interact with a subset of highly connected nodes [15]. Its potential with ecological implications is that nested networks often enhance community stability by promoting cooperative interactions [15]. Both the small and large networks showed "NA" when tested for nestedness, indicating that a clear hierarchical pattern of interactions was not detected. This result suggests that the interactions between bacteria and metabolites in these systems are not organised in such a way that specialists interact with subsets of the metabolites used by generalists, which is a hallmark of nested network structure. Instead, the interactions reflect a more competitive configuration, where bacteria are associated with specific metabolite sets rather than sharing a common hierarchy of resource use. The absence of nestedness is consistent with systems dominated by competitive dynamics, rather than mutualistic or cross-feeding relationships that typically promote nestedness [31]. As a result of this, these specific networks may stabilise through balanced competition, for example, rather than through the hierarchical overlap characteristic of nested ecological networks.

Overall, these network properties demonstrate that increasing system size enhances structural complexity, connectivity, and modularity, which may improve robustness through functional compartmentalisation. However, the absence of clustering and nestedness suggests limited redundancy: the loss of highly connected bacteria functioning as metabolic keystones cannot be easily compensated by other community members. This combination of modular organisation without nested redundancy

indicates that perturbations may be contained within modules, but critical species losses within modules could lead to irreversible functional shortfalls, aligning with the marginal stability observed in the eigenvalue analysis.

4.2 Implications for microbiota interventions

Metabolite concentrations and microbe levels in a system can be measured experimentally, however there are pitfalls to this method. Inferring microbial dynamics using common experimental data, particularly time-resolved metagenomics measurements, faces several critical pitfalls that impede accurate mathematical modelling. A fundamental challenge is the compositional nature of relative abundance data derived from sequencing, which is inherently problematic if the total microbial biomass of the ecosystem is time-variant, as this distortion prevents reliable representation of the original absolute dynamics [38]. Additionally, the temporal data collected is often not informative due to low sampling rates or insufficiently perturbing the system, which is crucial for parameter identification. When modelling highly complex systems like the gut microbiota, common practices intended to simplify the problem, such as ignoring or grouping low-abundance species, typically generate distorted interaction networks, even though the species may be strongly interacting [38].

The baseline results highlight the potential for manipulation of microbial–metabolite interactions, which has direct implications for microbiota-targeted interventions. Because metabolite dynamics are strongly shaped by the interaction coefficients ($c_{i,j}$) as shown in Figure 1, altering these parameters could shift the equilibrium levels of key metabolites in predictable ways. Some examples to alter these parameters could be through dietary changes, prebiotic supplementation, or engineered microbial strains. This manipulation potential is particularly relevant for therapeutic strategies, as controlling the production or depletion of certain metabolites may influence host physiology, gut health, and disease outcomes. For example, promoting the stability of beneficial metabolites while limiting the accumulation of harmful byproducts could help maintain a balanced microbial ecosystem. These findings suggest that even relatively simple models can provide insight into how targeted perturbations might be used to

steer microbial communities toward more desirable functional states, forming a foundation for rational microbiota-based interventions.

4.3 Limitations and future work

While the modelling approach provides valuable insights into the ecological behaviour of microbial communities, several limitations should be acknowledged, which also highlight directions for future work. A primary limitation lies in parameter uncertainty, where many of the model parameters are drawn from uniform distributions rather than estimated from empirical data. As a result, the model captures general trends in bacterial-metabolite dynamics but lacks quantitative precision [38]. In biological systems, these parameters are highly context-dependent and vary between species, environments, and host conditions. Therefore, the model's predictions should be interpreted qualitatively, as indicators of potential behaviours rather than precise forecasts of microbial abundances or metabolite concentrations.

Another limitation arises from the simplified representation of microbial interactions. The model assumes that the effects of metabolites on bacterial growth can be represented through linear and Hill-type functional forms. While these formulations are mathematically useful, they oversimplify the complex biochemical and ecological relationships that define microbial ecosystems. Real microbial communities exhibit diverse interaction modes, including cross-feeding, syntrophy, and competition for multiple metabolites, all of which are influenced by regulatory networks and environmental conditions [36]. By abstracting these processes into simplified interaction terms, the model omits important biological detail and nonlinear feedback mechanisms that could influence stability. Future extensions should expand the interaction network to include processes such as cross-feeding, syntrophy, and competitive inhibition to provide a more mechanistic understanding of coexistence and metabolic interdependence within microbial communities [17].

The deterministic framework of the model presents another limitation, as it represents all processes as continuous and predictable changes over time. This approach neglects stochastic effects such as random fluctuations in metabolite concentrations, bacterial abundance, and environmental variability [24]. Introducing stochastic differential equations or agent-based approaches in future work could help capture

random events and reveal alternative stable states, transient dynamics, or extinction events that deterministic models inherently overlook [16].

The model is also limited by its temporal and spatial resolution. It treats the gut environment as a well-mixed, homogeneous system, ignoring spatial gradients in nutrient concentrations, oxygen levels, and microbial distributions. In reality, the gut microbiome is spatially structured, with distinct niches and gradients that influence microbial competition and coexistence [24]. Including spatial heterogeneity, diffusion terms, time delays, or compartmental structures in future versions could capture how physical gradients and transport processes affect microbial behaviour, leading to a more realistic and mechanistically grounded representation of microbiome dynamics [23].

Finally, a useful direction for future work is the incorporation of empirical data to move beyond theoretical simulations. While the current framework demonstrates baseline dynamics and the influence of interaction coefficients on metabolite trajectories, fitting the model to experimental measurements would allow key parameters such as the interaction coefficients, metabolite turnover rates, and growth constants to be estimated directly from biological systems [22]. This step is critical for improving the predictive accuracy and biological relevance of the model. By calibrating simulations with real data, it will be possible to validate the model structure, quantify parameter uncertainty, and assess its capacity to reproduce observed microbial and metabolite dynamics. Ultimately, integrating empirical data would strengthen the model's utility for generating biologically meaningful insights and could support the design of targeted microbiota interventions by identifying the parameters that most strongly influence system behaviour [22].

In summary, addressing parameter uncertainty, extending the interaction framework, incorporating stochastic and spatial complexity, and grounding the model in empirical data represent key areas for future development. Together, these improvements would bridge the gap between theoretical modelling and biological reality, enabling a more accurate and mechanistically informed understanding of gut microbiome stability and resilience.

5 Conclusion

The gut environment is a remarkably complex and dynamic ecosystem shaped by intricate interactions between microbial species and the metabolites they produce and consume. Capturing this complexity through experimental observation alone is challenging, making mathematical modelling a powerful tool for understanding its underlying dynamics. By constructing and simulating mathematical models of bacterial-metabolite interactions, this study demonstrated how nonlinear behaviours, such as non-monotonic responses and feedback effects, can emerge naturally within microbial communities.

Extending this modelling framework to include stability analysis provided insights into how the system responds to perturbations and how different parameter regimes influence equilibrium states. Furthermore, by translating interaction data into network representations, we were able to examine structural properties such as connectivity, modularity, and centrality, each of which contributes to understanding the ecological organisation and resilience of the system.

Together, these approaches highlight the value of mathematical modelling as a helpful tool which can be used to explore the gut. Integrating dynamic simulations, stability assessment, and network theory advances our ability to interpret microbial interactions and provides a foundation for predicting how these ecosystems respond to environmental or therapeutic interventions. As modelling becomes increasingly coupled with experimental and experimental data, it will offer deeper, more predictive insights into the complex ecology of the gut microbiome.

6 References

1. Fujisaka, S., Y. Watanabe, and K. Tobe, *The gut microbiome: a core regulator of metabolism*. Journal of Endocrinology, 2023. **256**(3): p. e220111.
2. Rowland, I., et al., *Gut microbiota functions: metabolism of nutrients and other food components*. Eur J Nutr, 2018. **57**(1): p. 1-24.
3. Dalamaga, M. and C. Tsigalou, *Diet patterns, gut microbiota and metabolic disorders: Perspectives and challenges*. Metabol Open, 2024. **23**: p. 100310.
4. Nicholson, J.K., et al., *Host-gut microbiota metabolic interactions*. Science, 2012. **336**(6086): p. 1262-1267.
5. Sekirov, I., et al., *Gut Microbiota in Health and Disease*. Physiological Reviews, 2010. **90**(3): p. 859-904.
6. Hou, K., et al., *Microbiota in health and diseases*. Signal Transduction and Targeted Therapy, 2022. **7**(1): p. 135.
7. Liu, J., et al., *Functions of Gut Microbiota Metabolites, Current Status and Future Perspectives*. Aging Dis, 2022. **13**(4): p. 1106-1126.
8. Lin, L. and J. Zhang, *Role of intestinal microbiota and metabolites on gut homeostasis and human diseases*. BMC Immunology, 2017. **18**(1): p. 2.
9. Lozupone, C.A., et al., *Diversity, stability and resilience of the human gut microbiota*. Nature, 2012. **489**(7415): p. 220-230.
10. Olszak, T., et al., *Microbial exposure during early life has persistent effects on natural killer T cell function*. Science, 2012. **336**(6080): p. 489-93.
11. Cryan, J.F. and T.G. Dinan, *Mind-altering microorganisms: the impact of the gut microbiota on brain and behaviour*. Nature Reviews Neuroscience, 2012. **13**(10): p. 701-712.
12. Bajic, D. and A. Sanchez, *The ecology and evolution of microbial metabolic strategies*. Current Opinion in Biotechnology, 2020. **62**: p. 123-128.
13. Basan, M., et al., *Overflow metabolism in Escherichia coli results from efficient proteome allocation*. Nature, 2015. **528**(7580): p. 99-104.
14. Harcombe, W.R., et al., *Adding biotic complexity alters the metabolic benefits of mutualism*. Evolution, 2016. **70**(8): p. 1871-1881.
15. Oña, L., S.K. Shreekar, and C. Kost, *Disentangling microbial interaction networks*. Trends in Microbiology, 2025. **33**(6): p. 619-634.
16. Saa, P., et al., *Modeling approaches for probing cross-feeding interactions in the human gut microbiome*. Computational and Structural Biotechnology Journal, 2022. **20**: p. 79-89.
17. Bakkeren, E., V. Piskovsky, and K.R. Foster, *Metabolic ecology of microbiomes: Nutrient competition, host benefits, and community engineering*. Cell Host and Microbe, 2025. **33**(6): p. 790-807.
18. Plata, G., et al., *Designing host-associated microbiomes using the consumer/resource model*. mSystems, 2024. **10**(1): p. e01068-24.
19. Schlomann, B.H. and R. Parthasarathy, *Timescales of gut microbiome dynamics*. Curr Opin Microbiol, 2019. **50**: p. 56-63.
20. Liao, C., et al., *Modeling microbial cross-feeding at intermediate scale portrays community dynamics and species coexistence*. bioRxiv, 2020: p. 2020.02.19.956383.

21. Stein, R.R., et al., *Ecological Modeling from Time-Series Inference: Insight into Dynamics and Stability of Intestinal Microbiota*. PLOS Computational Biology, 2013. **9**(12): p. e1003388.
22. Bucci, V., et al., *MDSINE: Microbial Dynamical Systems INference Engine for microbiome time-series analyses*. Genome Biology, 2016. **17**(1): p. 121.
23. Cremer, J., M. Arnoldini, and T. Hwa, *Effect of water flow and chemical environment on microbiota growth and composition in the human colon*. Proc Natl Acad Sci U S A, 2017. **114**(25): p. 6438-6443.
24. Hellweger, F.L., et al., *Advancing microbial sciences by individual-based modelling*. Nat Rev Microbiol, 2016. **14**(7): p. 461-71.
25. Shoaie, S., et al., *Quantifying Diet-Induced Metabolic Changes of the Human Gut Microbiome*. Cell Metabolism, 2015. **22**(2): p. 320-331.
26. Magnúsdóttir, S., et al., *Generation of genome-scale metabolic reconstructions for 773 members of the human gut microbiota*. Nature Biotechnology, 2017. **35**(1): p. 81-89.
27. Kurtz, Z.D., et al., *Sparse and Compositionally Robust Inference of Microbial Ecological Networks*. PLoS Computational Biology, 2015. **11**(5).
28. Jansma, J., P. Landi, and C. Hui, *Bayesian inference captures metabolite-bacteria interactions in a microbial community*. bioRxiv, 2025: p. 2025.09.11.675530.
29. Nev, O.A., et al., *Predicting microbial growth dynamics in response to nutrient availability*. PLOS Computational Biology, 2021. **17**(3): p. e1008817.
30. Jansma, J., et al., *Metabolic network construction reveals probiotic-specific alterations in the metabolic activity of a synthetic small intestinal community*. mSystems, 2023. **8**(5): p. e00332-23.
31. Coyte, K.Z., J. Schluter, and K.R. Foster, *The ecology of the microbiome: Networks, competition, and stability*. Science, 2015. **350**(6261): p. 663-666.
32. Fassarella, M., et al., *Gut microbiome stability and resilience: elucidating the response to perturbations in order to modulate gut health*. Gut, 2021. **70**(3): p. 595.
33. Zhou, H., et al., *Exploring and evaluating microbiome resilience in the gut*. FEMS Microbiology Ecology, 2025. **101**(5): p. fiaf046.
34. Mizuno, K., et al., *Adjacent-possible ecological niche: growth of Lactobacillus species co-cultured with Escherichia coli in a synthetic minimal medium*. Scientific Reports, 2017. **7**(1): p. 12880.
35. Sung, J., et al., *Global metabolic interaction network of the human gut microbiota for context-specific community-scale analysis*. Nature Communications, 2017. **8**(1): p. 15393.
36. Faust, K. and J. Raes, *Microbial interactions: from networks to models*. Nature Reviews Microbiology, 2012. **10**(8): p. 538-550.
37. Levy, R. and E. Borenstein, *Metabolic modeling of species interaction in the human microbiome elucidates community-level assembly rules*. Proceedings of the National Academy of Sciences, 2013. **110**(31): p. 12804-12809.
38. Cao, H.T., et al., *Inferring human microbial dynamics from temporal metagenomics data: Pitfalls and lessons*. Bioessays, 2017. **39**(2).

7 Appendix

All code used in this study is publicly available at:

https://github.com/jemmabennett/Honours-Project-Gut-Microbiota-Modelling.git

Figure 1: Simulations of a single run

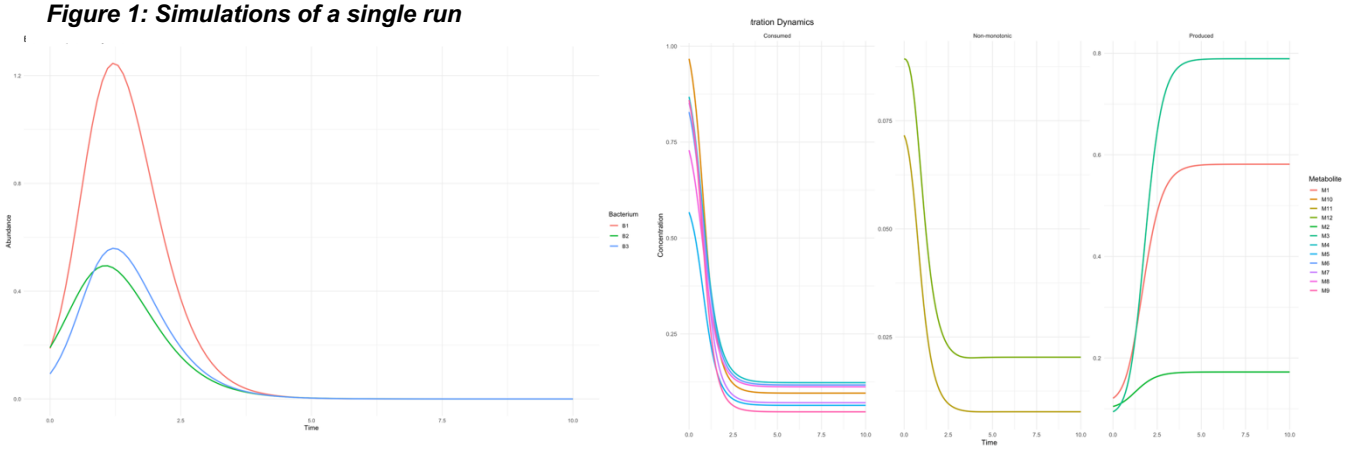

Figure 2: Example of values used in simulations drawn from uniform distributions

```
> c_matrix
            [,1]       [,2]        [,3]
 [1,] -0.08519396 -0.1695524 -0.26341169
 [2,] -0.06292459 -0.3582545 -0.86533340
 [3,] -0.71386047 -0.4809041 -0.34300771
 [4,]  0.70506478  0.9782264  0.94666823
 [5,]  0.45774178  0.1174874  0.08243756
 [6,]  0.71911225  0.4749971  0.51421178
 [7,]  0.93467225  0.5603327  0.39020347
 [8,]  0.25542882  0.9040314  0.90573813
 [9,]  0.46229282  0.1387102  0.44696963
[10,]  0.94001452  0.9888917  0.83600426
> c_prod
           [,1]         [,2]       [,3]
[1,] -0.3277838 -0.519382264 -0.5492260
[2,] -0.3545496 -0.013481417 -0.2851512
[3,] -0.7917145 -0.006880152 -0.6834384
> c_cons
          [,1]      [,2]      [,3]
[1,] 0.8336876 0.0520156 0.4779508
[2,] 0.4339756 0.4587231 0.7688522
[3,] 0.9959923 0.4981566 0.1639289
[4,] 0.8058112 0.2827642 0.4417741
[5,] 0.4862422 0.7764451 0.5234077
[6,] 0.5387720 0.3038528 0.4769192
[7,] 0.3410399 0.5155512 0.5246433
> c_nonmono_L
           [,1]        [,2]       [,3]
[1,] -0.3147718 -0.16036871 -0.8936904
[2,] -0.6482080 -0.07808631 -0.7934711
> c_nonmono_U
          [,1]      [,2]      [,3]
[1,] 0.4866429 0.4717139 0.2239500
[2,] 0.3238220 0.5860885 0.7340297
> w
          [,1]      [,2]      [,3]
[1,] 0.2354357 0.3376971 0.7492002
[2,] 0.7277980 0.4838184 0.5270245
>
```

Figure 3: Summary of open, closed, large and small system simulations

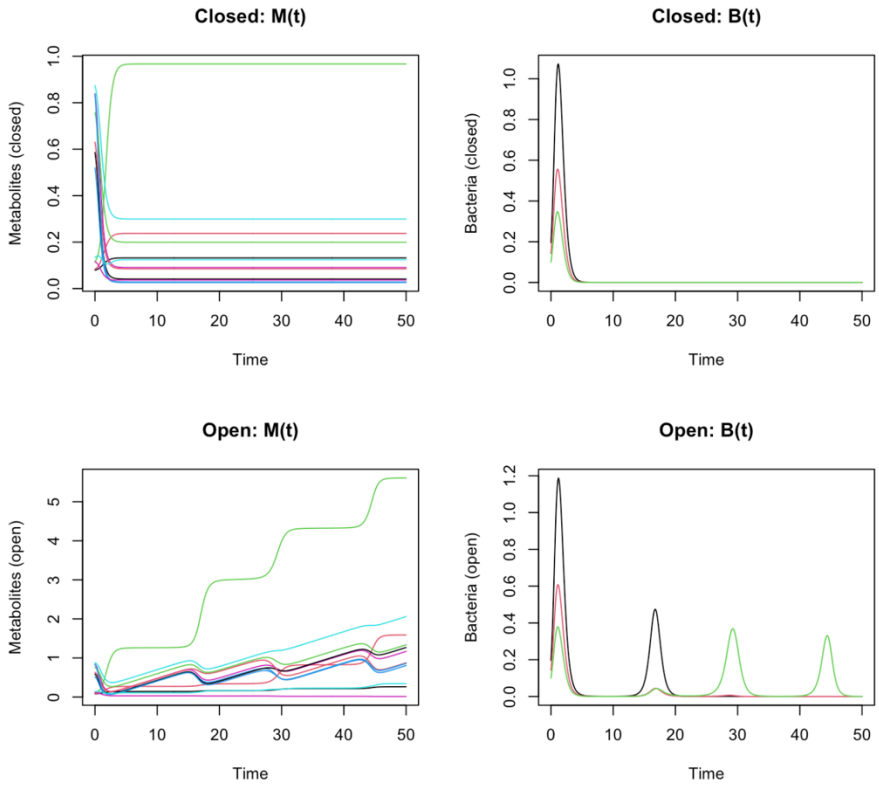

Figure 4: Bifurcation diagrams of non-monotonic metabolites

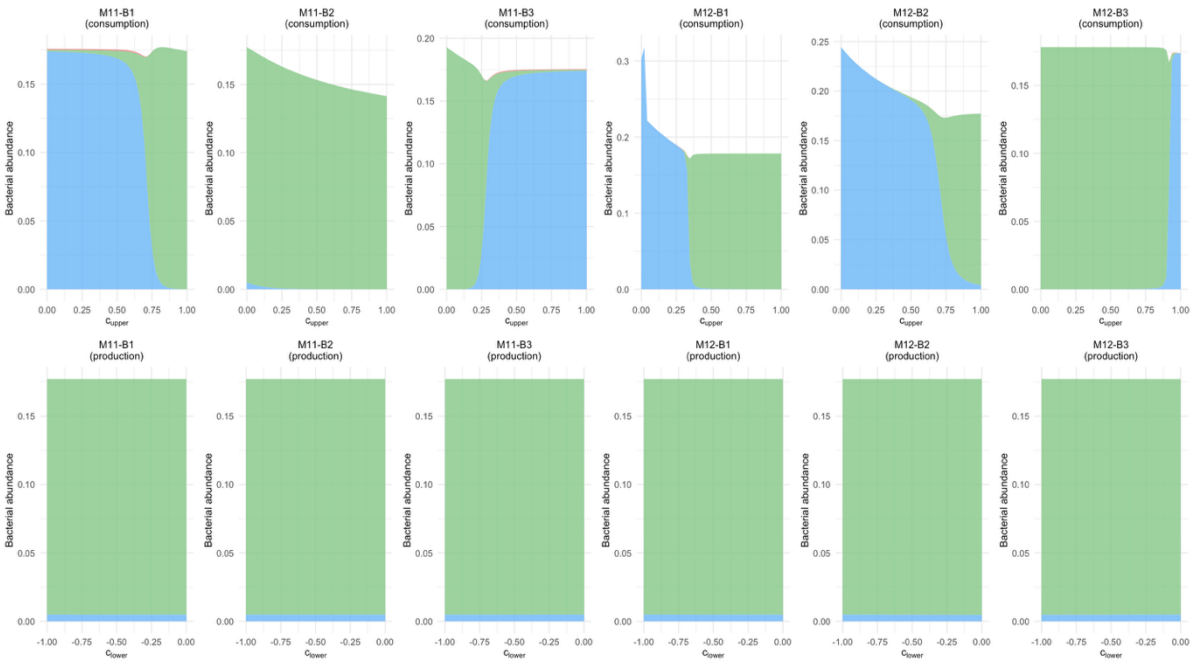

Figure 5: Interaction network at time t = 20 of small network

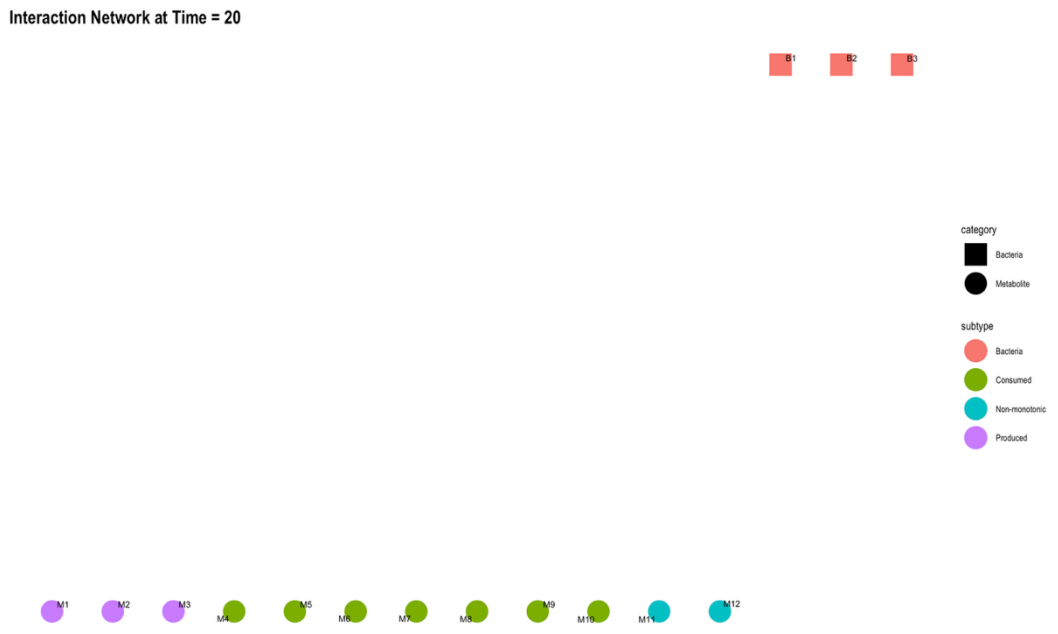
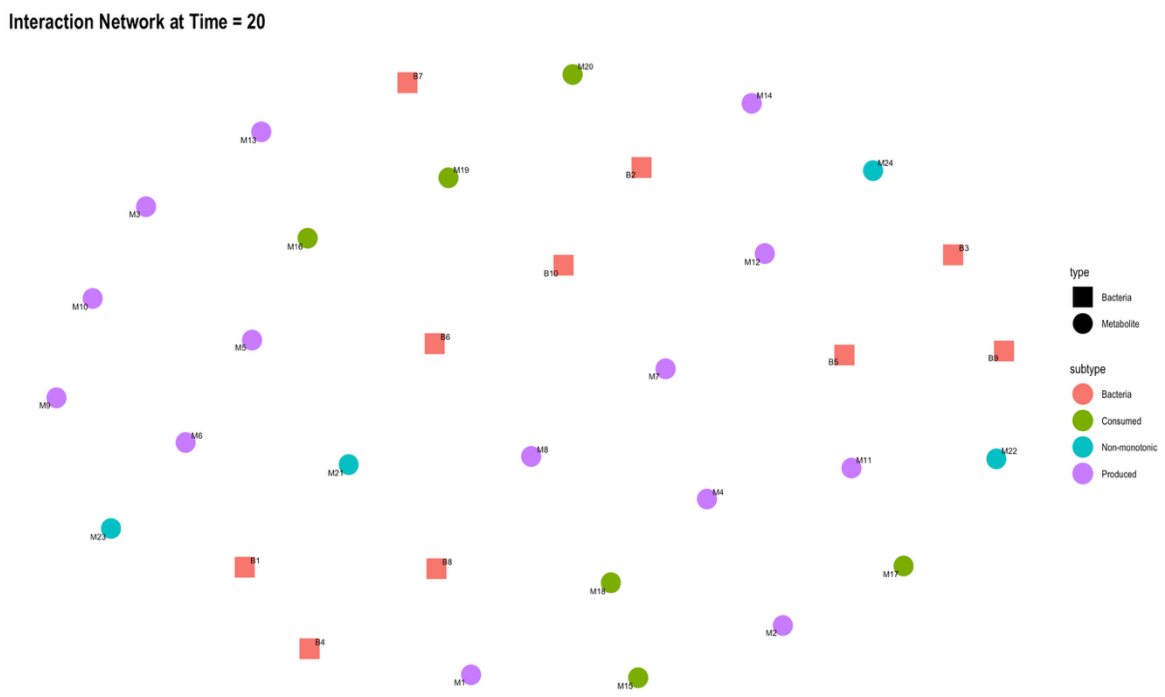
Figure 6: Interaction network at time t = 20 of large network